Informatica® Business 360 Console
October 2026

Configure Match and Merge

Informatica Business 360 Console Configure Match and Merge
October 2026

© Copyright Informatica LLC 2020, 2026

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html. Other company and product names may be trade names or trademarks of their respective owners.

Portions of this software and/or documentation are subject to copyright held by third parties. Required third party notices and AI assistant acknowledgements are included with the product.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-10-16

Table of Contents

Preface 6

Informatica Resources 6

- Informatica Documentation 6
- IDMC web site 6
- Intelligent Data Management Cloud Communities 6
- IDMC Marketplace 6
- Informatica Knowledge Base 7
- IDMC Trust Center 7
- Informatica Global Customer Support 7

Chapter 1: Configuring match and merge 8

Prerequisites for configuring match and merge 8

Match and merge statuses 9

Chapter 2: Match model properties 12

Model name 12

Model objective 12

Default Population 13

- Supported Populations 13

Candidate selection criteria 19

- Field name 20
- Field type 20
- Filter candidates 22
- Key generation level 22
- Candidate search level 23
- Scenarios for candidate selection criteria configuration 24
- Guidelines for configuring candidate selection criteria 25

Maximum candidates 25

Chapter 3: Configuring candidate selection criteria 26

Add candidate selection criteria 27

Managing candidate selection criteria 27

- Enable candidate selection criteria 28
- Disable candidate selection criteria 28
- Delete candidate selection criteria 28

Chapter 4: Configuring match models 29

Creating a match model from a copy 29

- Step 1. Review the existing match model 29
- Step 2. Create a copy of a match model 30

Step 3. Configure a match model. . . . 30
Creating a match model from scratch. . . . 31
Editing a match model. . . . 32
Deleting a match model. . . . 33
Distinct source systems. . . . 33
Configuring a distinct source system. . . . 34

Chapter 5: Directed AI match rule properties. . . . 35

Description. . . . 35
Match strategy. . . . 36
Merge strategy. . . . 36
Merge threshold. . . . 37
Match criteria. . . . 37
Match level. . . . 39
Rank. . . . 39
Match fields. . . . 41
Field type. . . . 42
Additional required fields and optional fields. . . . 43
Field properties. . . . 43
Segment matching. . . . 43
Null matching. . . . 47
Radius. . . . 49
Guidelines for configuring Directed AI match rules. . . . 49

Chapter 6: Configuring Directed AI match rules. . . . 50

Step 1. Review the data. . . . 50
Step 2. Add Directed AI match rules. . . . 51
Step 1. Configure match properties and threshold. . . . 51
Step 2. Add required fields. . . . 52
Step 3. Add optional fields. . . . 53
Step 4. Configure field properties. . . . 53
Step 3. Set the execution order for Directed AI match rules. . . . 54
Edit Directed AI match rules. . . . 54

Chapter 7: Configuring Directed AI match rules with relationships. . . . 56

Guidelines for configuring Directed AI match rules with relationships. . . . 56
Configure Directed AI match rules with relationship match criteria. . . . 57

Chapter 8: Training adaptive AI match models. . . . 58

Training metrics. . . . 58
Error matrix. . . . 59
Accuracy. . . . 59
Precision. . . . 60

Recall. . . . 60
Learning curve. . . . 60
Recommendations for selecting training data sets. . . . 61
Recommendations for training adaptive AI match models. . . . 62
Prepare to train an Adaptive AI match model. . . . 62
Step 1. Select a training data set. . . . 62
Step 2. Import the training data set. . . . 63
Step 3. Identify a match model. . . . 64
Step 4. Configure match fields. . . . 64
Training the Adaptive AI match model. . . . 65
Label record pairs for training an Adaptive AI match model. . . . 66
Reset an Adaptive AI match model training. . . . 69
Keyboard shortcuts. . . . 72

Chapter 9: Publishing match models. . . . 73

Publish a match model. . . . 73
Additional training for Adaptive AI match models. . . . 74
Continue training an Adaptive AI match model. . . . 74
Retrain an Adaptive AI match model. . . . 77
Retrain as a new version. . . . 77

Chapter 10: Setting the default match model for the search match API. . . . 80

Index. . . . 81

Preface

Follow the instructions in *Configure Match and Merge* to configure the match and merge process, and identify and resolve duplicate records. Learn how to create match models and configure match rules.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

Intelligent Data Management Cloud Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

To access the IDMC Community, see Master Data Management SaaS Community.

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Configuring match and merge

Configure the match and merge process to identify and resolve duplicate records. The match and merge process configuration together constitutes the match model. Based on the outcome of the match process and the merge configuration, the records are either automatically merged, queued for manual merge by a data steward, or skipped as not a candidate for merge.

To resolve duplicates, you can choose to use an Adaptive AI match model, Directed AI match rules, or both. The Directed AI match rules are based on an exact or fuzzy match strategy. An exact match strategy identifies identical records, and a fuzzy match strategy identifies similar records. To reduce overmatching and provide a better matching performance, consider adding at least one exact match rule in a match model.

An Adaptive AI match model needs to undergo a training process based on your business needs, where a user labels record pairs to train the Adaptive AI match model. An Adaptive AI match model training reduces the manual effort required for defining Directed AI match rules.

To configure the match and merge process, use the predefined match model or create a custom model to suit your business needs. You can create a custom match model from scratch or from a copy of an existing match model. You cannot edit the predefined model.

Important: You must configure and publish the match model after you configure the data model, but before the ingress job is run for the initial data load. If the match model configuration is published after the initial data load, the data that is already in the Business 360 data store is not considered by the match model and does not undergo matching.

After you publish a model, you can define match and merge jobs, and run the jobs. For information about defining and running match and merge jobs, see *Define Jobs*.

In addition to configuring match and merge rules, configure survivorship to resolve conflicting data from matched records. You can specify which values to retain for specific attributes when multiple records are merged.

Prerequisites for configuring match and merge

Before you configure the match and merge process, analyze your data and identify its key characteristics. Also, understand the specific requirements for match and merge. Meaningful insights into your data can help you make better decisions when you configure Directed AI match rules.

Ensure that you understand the business objectives and requirements for matching and merging duplicate data. You can also analyze the attributes and quality of data and identify anomalies and patterns that you want to consider for the match and merge process.

The following table describes different tasks that you can perform to analyze data:

Task	Description
Analyze data	Analyze the data samples and identify the fields that you can use for matching. You can use SQL queries, data profiler, or other tools to analyze data. When you analyze the data samples, you can identify the following details: - Identify the most critical attributes that hold unique identifiers, such as phone numbers and social security numbers, and distinct patterns, such as postal codes and email addresses. - Determine the size of the data set, which could impact the performance of the match and merge process. - Determine the match population, which improves match accuracy by accommodating variations and errors that are likely to appear in data for a particular population. Additionally, using a data profiler or other tools helps you identify granular details, such as the cardinality that exists between fields, data completeness and uniqueness of fields.
Collect data samples	Collect a representative data sample from various sources, excluding test system data, and ensure its size is manageable for analysis.
Determine match requirements	Talk to users to determine the high-level match requirements and understand what qualifies as a match based on the sample data set.
Identify groups of similar records	Identify similar records by running simple group statistics on single or multiple field data. The group statistics help you identify the following details: - Large groups of identical records. - Appropriate candidate selection criteria. A large number of candidates might impact match performance. - Determine viability of the intended match key fields.
Verify data completeness	Determine how often a field has a non-null value and look for data completeness by fields and combination of fields. For example, determine the percentage of records that have both first and last name values. If you want to use a field, such as postal code as an exact match field, the data set should contain more than 50% values for this field.
Determine data quality	Ensure that the data is accurate. For example, gender field must contain only gender values. You can use data quality rule associations to refine the data. You can also use pattern analysis to assess the quality of data in a field. Pattern analysis is useful in analyzing data that conforms to certain formats or data types, such as postal codes or email addresses.
Remove extraneous data	Identify and remove extraneous data by reviewing data manually or by using a data profiling tool.

Match and merge statuses

MDM SaaS assigns a match and merge status to each source record after you ingress it. The status is updated as the records progress through the match and merge process.

During the ingress process, the status of the records is first set to MATCH DIRTY. After the match keys are generated, the status of the records changes to MATCH INDEXED. All the records in the MATCH INDEXED status can go through the match process. The match process creates record pairs, and the status of the records changes to MATCHED. All the records in the MATCHED status can go through the merge process. After the records are automatically or manually merged, the status of these records changes to CONSOLIDATED.

If the records aren't considered as a match during a manual review, the status of the records remains as MATCHED. If the records don't match any of the existing records, the status of the records changes from MATCH INDEXED to MATCHED.

An updated record can participate in the match process only if you update the match fields. When you update the record fields that are configured as match fields in the business entity, the match and merge status of the record changes to MATCH DIRTY. When you update non-match fields, the match and merge status of the record doesn't change to MATCH DIRTY.

If you haven't merged the records and reset the MATCHED status of record pairs, the status of these records changes to MATCH INDEXED.

During the ingress process, you can also exclude a source record from participating in the match process by specifying it as not ready for match. The status of these records changes to NOT READY FOR MATCH, and they don’t participate in the match process. These records remain as unique master records until they are marked as ready for matching.

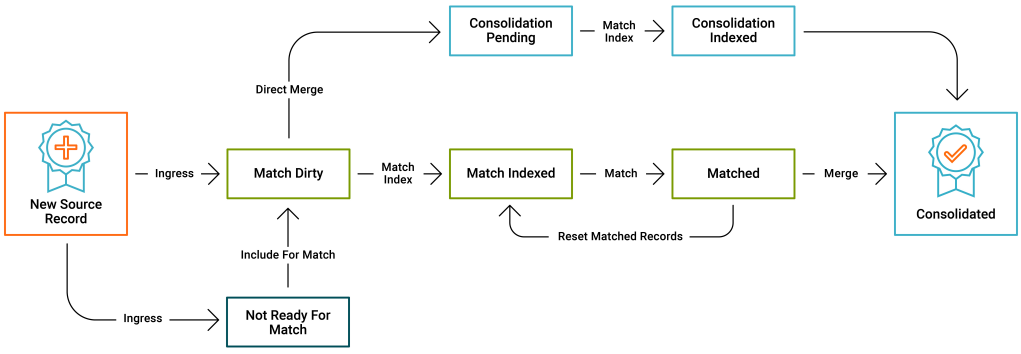
When you ingress source records with business IDs of existing records, the records don’t go through the match process. The status of the records changes from MATCH DIRTY to CONSOLIDATION PENDING. The ingress process generates match keys for records with the CONSOLIDATION PENDING status and changes the status to CONSOLIDATION INDEXED. The merge process merges the master records based on the business IDs specified during the ingress process, and the status of the records changes to CONSOLIDATED.

When you run a match and merge job, the records in the MATCH INDEXED status match with the records in the MATCHED or CONSOLIDATED status. However, the records in the MATCHED or CONSOLIDATED status don’t match with each other.

Effective in the May 2025 release, the ability for the match process to set the match and merge status of source records that don't match with any records to CONSOLIDATED is available for preview. The status of these source records is set to CONSOLIDATED only when the source record count is 1000 or more. Otherwise, the status of these records remains as MATCHED.

Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

The following image shows the different match and merge statuses of a source record:



The following table describes the different match and merge statuses of source records:

Match and Merge Status	Description
MATCH DIRTY	Indicates that the source records are ingressed but haven't undergone the match process.
MATCH INDEXED	Indicates that match keys are generated for the source records.
MATCHED	Indicates that the source records have matched with other records based on the match configuration, and they are ready to be merged.
CONSOLIDATION PENDING	Indicates that the source records haven't undergone the match process because the records were ingressed with business IDs of existing records.
CONSOLIDATION INDEXED	Indicates that match keys are generated for the source records ingressed with business IDs of existing records.
CONSOLIDATED	Indicates that the source records are merged either manually or automatically.
NOT READY FOR MATCH	Indicates that you've excluded the source records from participating in the match process.

Note: When you delete source records, MDM SaaS doesn’t permanently delete them from the data store. The state of these records changes from active to deleted in the data store. MDM SaaS changes the match and merge status of these records to MATCH DIRTY irrespective of their previous state.

CHAPTER 2

Match model properties

A match model is a set of properties or configuration that determines how your data is matched. Configure a match model based on the characteristics of your data and your business needs.

A match model includes the following properties:

- “Model name” on page 12
- “Model objective” on page 12
- “Default Population” on page 13
- “Candidate selection criteria” on page 19
- “Maximum candidates” on page 25

Match models include a set of Directed AI match rules and an Adaptive AI match model. The match process first uses the Directed AI match rules to match records. If the Directed AI match rules provide conclusive results, the match process doesn’t use the Adaptive AI match model. However, if the Directed AI match rules provide inconclusive results that require manual merge, the match process then uses the Adaptive AI match model on those records to determine whether they match.

For information about Directed AI match rules, see Chapter 5, “Directed AI match rule properties” on page 35. For information about adaptive AI match models, see Chapter 8, “Training adaptive AI match models” on page 58.

Model name

The model name helps identify the match model that you configure. This name is listed in the table on the **Match Model** page.

Note: You can't change the model name after the model is published.

Model objective

The model objective is the goal or problem that you want to solve by using a match model. You can use the Resolve Duplicates objective to identify similar and identical records to resolve duplicates.

The current release includes only the Resolve Duplicates objective and this is the default objective of a match model.

Default Population

Population defines specific characteristics of data in the records that you want to match. Each supported country, language, or population has a match population.

Populations improve match accuracy by accounting for the inevitable variations and errors that might exist in the name, address, and other identification data of a particular population. Populations contain the logic to generate match keys and specify how match strategies and match criteria process the data to be matched. You can analyze your data and determine the appropriate population to use.

You can use multiple populations for data that represents diverse populations, but when you ingress or import records, you can assign multiple populations to the records. You can't update the population of a record in business applications. If the records don't contain population information, the population that's configured for the match model is used. For example, your data consists of 25 records, with 15 records assigned to the Japan population and 10 records that don't have any specified population. Then, the default population of the match model, such as Usa, is applied to the 10 unspecified records during the matching process.

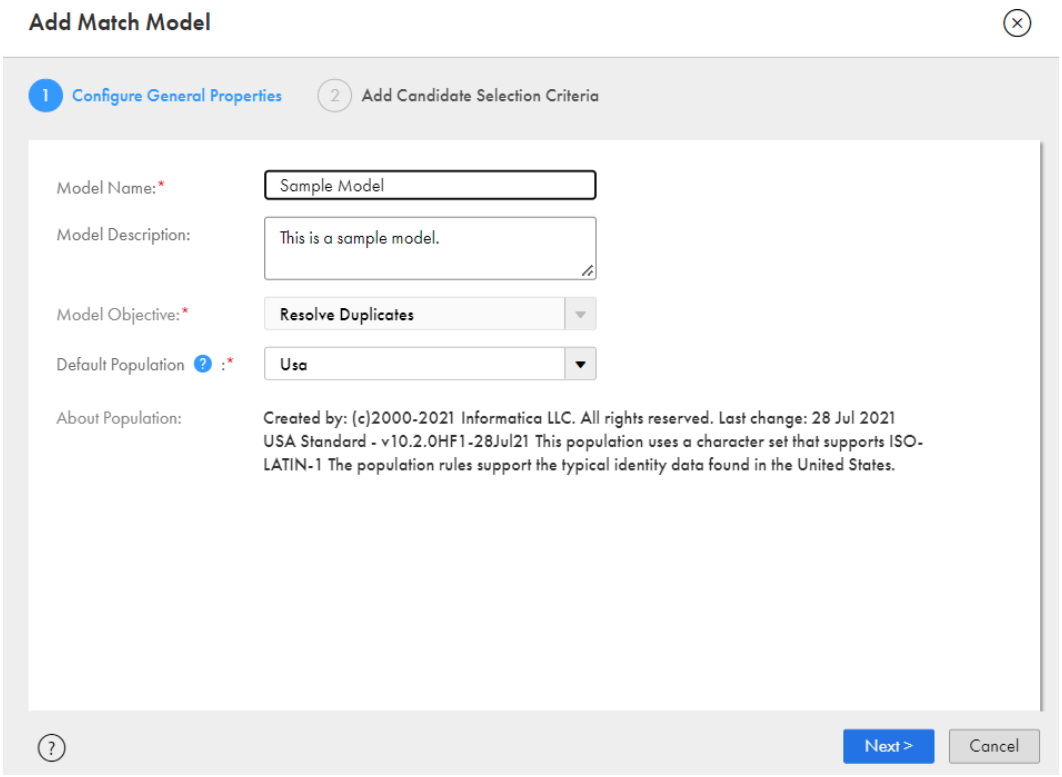
The match criteria values in a directed AI match rule vary based on the default match population that you select.

Supported Populations

You can select a default population from a list of values for different countries when you configure a match model.

You can view detailed information about the population. For example, if you select the Chinese_r population, you can view the details about the Chinese Romanized Standard script for the Chinese population.

The following image shows the details for the Usa population:



The following table lists the supported populations that you can use for a match model:

Population	Description	Unicode Support	Cross Script
Arabic	Contains the Arabic Standard script for the Arabia population.	No	No
Arabic_m	Contains a mix of Arabic, Romanized Arabic, English, and Arabicized English scripts for the Arabia population.	Yes	Yes
Arabic_r	Contains the Arabic Romanized Standard script for the Arabia population.	Yes	No
Argentina	Contains the Argentina Standard script for the Argentina population.	Yes	Not applicable
Australia	Contains the Australia Standard script for the Australia population.	Yes	Not applicable
Belgium	Contains the Belgium Standard script for the Belgium population.	Yes	Not applicable

Population	Description	Unicode Support	Cross Script
Brasil	Contains the Brasil Standard script for the Brasil population.	Yes	Not applicable
Bulgaria_c	Contains the Bulgarian Cyrillic Standard script for the Bulgaria population.	Yes	No
Bulgaria_r	Contains the Bulgarian Romanised script for the Bulgaria population.	Yes	Not applicable
Canada	Contains the Canada Standard script for the Canada population.	Yes	Not applicable
Chile	Contains the Chile Standard script for the Chile population.	Yes	Not applicable
Chinese	Contains the Chinese Standard script for the Chinese population.	Yes	Yes
Chinese_i	Contains the Traditional Chinese Standard script for the Chinese population.	Yes	Yes
Chinese_r	Contains the Chinese Romanised Standard script for the Chinese population.	Yes	Not applicable
Chinese_s	Contains the Simplified Chinese Standard script for the Chinese population.	Yes	Yes
Chinese_t	Contains the Traditional Chinese Standard script for the Chinese population.	Yes	Yes
Colombia	Contains the Colombia Standard script for the Colombia population.	Yes	Not applicable
Croatia	Contains the Croatia Standard script for the Croatia population.	Yes	Not applicable
Czech	Contains the Czech Standard script for the Czech population.	Yes	Not applicable
Denmark	Contains the Denmark Standard script for the Denmark population.	Yes	Not applicable
Estonia	Contains the Estonia Standard script for the Estonia population.	Yes	Not applicable

Population	Description	Unicode Support	Cross Script
Finland	Contains the Finland Standard script for the Finland population.	Yes	Not applicable
France	Contains the France Standard script for the France population.	Yes	Not applicable
Gaelic	Contains the Gaelic Standard script for the Gaelic population.	Yes	Not applicable
Germany	Contains the Germany Standard script for the Germany population.	Yes	Not applicable
Greek	Contains the Greek Standard script for the Greek population.	Yes	No
Greek_l	Contains the Greek Mixed Standard script for the Greek population.	Yes	No
Hebrew	Contains the Hebrew Standard script for the Hebrew population.	Yes	No
Hk_r	Contains the Hong Kong Romanized Standard script for the Hong Kong population.	Yes	Not applicable
Hungary	Contains the Hungary Standard script for the Hungary population.	Yes	Not applicable
India	Contains the India Standard script for the India population.	Yes	Not applicable
Indo_chin_r	Contains the Indo-China Standard Standard script for the Indo-China population.	Yes	Not applicable
Indonesia	Contains the Indonesia Standard script for the Indonesia population.	Yes	Not applicable
International	Contains the International Latin-1 Standard script.	Yes	Not applicable
Ireland	Contains the Ireland Standard script for the Ireland population.	Yes	Not applicable
Italy	Contains the Italy Standard script for the Italy population.	Yes	Not applicable
Japan	Contains the Japan Standard script for the Japan population.	Yes	Yes

Population	Description	Unicode Support	Cross Script
Japan_r	Contains the Japanese Romanized Standard script for the Japan population.	Yes	Not applicable
Kazakhstan	Contains the Kazakhstan Standard script for the Kazakhstan population.	Yes	Yes
Korean	Contains the Korean Standard script for the Korean population.	Yes	Yes
Korean_r	Contains the Korean Romanized Standard script for the Korean population.	Yes	Not applicable
Luxembourg	Contains the Luxembourg Standard script for the Luxembourg population.	Yes	Not applicable
Malaysia	Contains the Malaysia Standard script for the Malaysia population.	Yes	Not applicable
Malta	Contains the Malta Standard script for the Malta population.	Yes	Not applicable
Mexico	Contains the Mexico Standard script for the Mexico population.	Yes	Not applicable
Netherlands	Contains the Netherlands Standard script for the Netherlands population.	Yes	Not applicable
New Zealand	Contains the New Zealand Standard script for the New Zealand population.	Yes	Not applicable
Norway	Contains the Norway Standard script for the Norway population.	Yes	Not applicable
Peru	Contains the Peru Standard script for the Peru population.	Yes	Not applicable
Philippines	Contains the Philippines Standard script for the Philippines population.	Yes	Not applicable
Poland	Contains the Poland Standard script for the Poland population.	Yes	Not applicable
Portugal	Contains the Portugal Standard script for the Portugal population.	Yes	Not applicable

Population	Description	Unicode Support	Cross Script
Puerto_rico	Contains the Puerto Rico Standard script for the Puerto Rico population.	Yes	Not applicable
Romania	Contains the Romania Standard script for the Romania population.	Yes	Not applicable
Russia	Contains the Russia Standard script for the Russia population.	Yes	Yes
Singapore	Contains the Singapore Standard script for the Singapore population.	Yes	Not applicable
Slovakia	Contains the Slovakia Standard script for the Slovakia population.	Yes	Not applicable
South_africa	Contains the South Africa Standard script for the South Africa population.	Yes	Not applicable
Spain	Contains the Spain Standard script for the Spain population.	Yes	Not applicable
Sweden	Contains the Sweden Standard script for the Sweden population.	Yes	Not applicable
Switzerland	Contains the Switzerland Standard script for the Switzerland population.	Yes	Not applicable
Taiwan_r	Contains the Taiwan Romanised Standard script for the Taiwan population.	Yes	Not applicable
Thai	Contains the Thai Standard script for the Thailand population.	Yes	No
Thai_r	Contains the Thai Romanised Standard script for the Thailand population.	Yes	Not applicable
Turkey	Contains the Turkey Standard script for the Turkey population.	Yes	Not applicable
Uk	Contains the UK Standard script for the UK population.	Yes	Not applicable

Population	Description	Unicode Support	Cross Script
Usa	Contains the USA Standard script for the USA population.	Yes	Not applicable
Vietnam_r	Contains the Vietnamese Romanized Standard script for the Vietnam population.	Yes	Not applicable

The population details such as population code and description are stored in a reference data file. To view the population details, navigate to **Explore** > **Reference360** > **PopulationName** > **Values**.

Note: The system matches similar records that belong to the same population even though the records might be in different languages.

Candidate selection criteria

Match candidates are record pairs that are possible matches. The criteria for selecting match candidates depends on business entity fields.

Configure candidate selection criteria that are absolutely necessary to enhance the quality of match candidates. Having multiple criteria improves quality but affects the performance of the candidate selection process.

When you create a match model, the match process recommends match fields from the candidate selection criteria and Directed AI match rules for inclusion in the Adaptive AI match model training. You can modify the match model configuration even after you start training its Adaptive AI match model. You can add or remove match fields from existing Directed AI match rules, or add new Directed AI match rules to the match model. The match process then displays the newly added match fields as recommendations for inclusion in the Adaptive AI match model training.

Additionally, you need to retrain the model if you modify the following settings of a match model during or after training the Adaptive AI match model:

- Default population
- Candidate selection criteria
- Match fields in the Adaptive AI match model

A candidate selection criteria consists of the following parameters:

- “Field name” on page 20
- “Field type” on page 42
- “Filter candidates” on page 22
- “Key generation level” on page 22
- “Candidate search level” on page 23

Note: When you modify the default population and the parameters in the candidate selection criteria section, ensure that you publish and regenerate the match keys for the records.

For more information about publishing and regenerating match keys, see “Publish a match model” on page 73.

Field name

Field name is the name of the business entity field to use for generating match keys, and then identifying match candidates.

For example, you can select a business entity field, such as Full Name. When you select a field, you need to select a corresponding field type to indicate the type of data the field contains.

Field type

A field type indicates the type of data contained in the business entity field that you select as your candidate selection criteria. The fuzzy matching process is effective only when you configure an appropriate field type for the business entity fields.

The following table describes the field types:

Field Type	Description
Address_Part1	Contains address details, such as building name, street number, street name, street type, and apartment details.
Code	Contains numeric or alphanumeric values, such as license number or vehicle registration number.
CreditCard	Contains credit card numbers.
Date	Contains dates, such as date of birth, expiry date, date of contract, date of change, or issued date.
Email	Contains email addresses.
Geocode	Contains geographic coordinates, such as latitude, longitude, and elevation. For more information about the Geocode field type, see "Geocode field type" on page 21.
ISBN10	Contains the 10-digit International Standard Book Number (ISBN).
ISBN13	Contains the 13-digit ISBN.
Organization_Name| Company_Name	Contains organization names, such as business names, institution names, department names, agency names, or trading names.
Person_Name	Contains names of individuals.
Product_Description	Contains description of products.
Product_Name	Contains names of products.
Telephone_Number	Contains telephone numbers.
VIN	Contains Vehicle Identification Numbers (VINs).

Geocode field type

You can match addresses based on the geographic coordinates instead of relying only on the textual address data for matching. The match process calculates the distance between the addresses based on the latitude, longitude, and elevation to determine the proximity.

Based on the proximity of the addresses, the match process considers them either as a match or not a match. You can either manually specify geographic coordinates in the addresses of records or use the enriched address values from the address verification DaaS provider. For more information about address verification, see *DaaS rule association for real-time processing.*

For example, you work for a telecommunication company that aims to group customer records based on the location of their customer care centers to enhance customer support. Based on the number of customers around each customer care center, you want to adjust the number of support personnel. To achieve this goal, you can identify customer addresses that fall within a radius of 1000 meters from their customer care centers based on the geographic coordinates.

To match records based on geographic coordinates, ensure that you configure a match model that uses the Geocode fields, latitude, longitude, and elevation in the candidate selection criteria and directed AI match rules. The Geocode field type contains the Latitude, Longitude, and Elevation fields. When you add the Geocode field type as the candidate selection criteria, the match process adds the Latitude, Longitude, and Elevation fields from the Address field group by default. If required, you can specify other business entity fields as the Latitude, Longitude, and Elevation fields. When you specify these fields, ensure that they belong to the same field group or they are root fields.

When you use the Geocode field type as candidate selection criteria, include other field types as additional candidate selection criteria to identify good candidates. You can use these fields to match addresses based on their proximity only when they are configured in Directed AI match rules.

Note: When you add the Geocode fields as match fields to train an Adaptive AI match model, the model treats them as regular text and doesn't use the geographic coordinates in the fields to match addresses by proximity.

Latitude and longitude fields

You can specify the latitude and longitude field values in the Decimal Degrees (D.D°) or Degrees Minutes Seconds (D° M' S") format. For example, you can specify the values either as 13.0843° N or 13°05'3.48" N. Additionally, you can use the following variations when you specify these values:

- 13.0843° N
- 13.0843N
- 13.0843 N
- N13.0843
- 13.0843
- -13.0843
- 13d05’3.48”
- 13d5’3.48”N
- 13:5:3.48N
- 13:5:3.48

If directional indicators, such as N, S, E, and W, aren't added to indicate north, south, east, or west, the match process uses north and east as the default latitude and longitude values. Additionally, the match process considers positive and negative latitude values as north and south. Similarly, positive and negative longitude values are considered as east and west.

Elevation field

You can specify a numerical value for the elevation field. For example, you can specify the value as 100m. Additionally, you can use the following variations when you specify a value:

- 100m
- 100ft
- 100km
- 100miles
- 100

The default unit is meter. If you don't configure the Elevation field, the match process excludes the field from matching.

Radius

You can specify a radius to determine the proximity of addresses. To avoid missing any potential matches, the addresses of match candidates might also fall outside the specified radius. The value can be between 1 to 10,000 meters. Default is 1000.

Filter candidates

You can use filters to identify potential matches within a subset of data based on specified criteria. When you add a filter, you might have a fewer number of match candidates resulting in improved performance and accuracy of matching.

For example, if you select Full Name as the candidate selection field and Country as the filter field, the candidate selection process filters match candidates with similar full names for each country. If the records have country values of Germany and France, when you match a record with the country value Germany, the candidate selection process matches it with other records that also have country value as Germany, rather than matching it with all records. This process generates fewer candidates and the performance of matching improves.

Key generation level

The key generation level defines the thoroughness with which the match model generates match keys to identify candidates for matching. The key generation levels are extended, limited, and standard.

Decide on a key generation level based on the following considerations:

- Size and quality of the data
- Reliability of the matched records
- Processing time for key generation

The following table describes the key generation levels:

Key Generation Level	Description
Extended	Performs a stringent analysis of your data to generate many match keys, which might result in a large number of match candidates. This level ensures reliable matches, but at the cost of long processing time to generate match keys. If your data volume is not too large and the data quality is poor, use the extended key generation level.
Limited	Performs a quick and lenient analysis of your data to generate few match keys, which might result in few match candidates. This level trades some match reliability for a faster search of match candidates compared to when you use the extended key generation level. If your data volume is extremely large with variations in word order, use the limited key generation level. The results of the limited key generation level are a subset of the standard level.
Standard	Generates an optimum number of match keys, which in turn results in an optimum number of match candidates. This level balances the reliability of matches and the processing time to generate match keys.

Candidate search level

The match model uses the candidate search level to determine how stringently and thoroughly to search for match candidates. Later, the match model applies the Directed AI match rules on the match candidates and not on the entire data. You can configure an exhaustive, extreme, narrow, or typical candidate search level.

The match model uses the key generation level and the candidate search level that you define to determine the records that are potential match candidates.

Decide on a candidate search level based on the following considerations:

- Size and quality of the data
- Criticality of the matches
- Time constraints

The following table describes the candidate search levels:

Candidate Search Level	Description	Processing Time	When to Use
Extreme	Performs a lenient search that returns many more match candidates than the exhaustive search level. This search level results in a larger number of match candidates than the other search levels, possibly resulting in overmatching, and processes slowly.	Highest	Appropriate for small volumes of data that aren't not reliable and incomplete, or if it is critical to find the highest possible number of match candidates. You can also use the extreme level for the initial phase of matching and use a more stringent search level for the later phase.
Exhaustive	Performs a lenient search that returns many match candidates. This search level can result in more match candidates than most other search levels, possibly resulting in overmatching, and processes slowly.	High	Appropriate for small volumes of data that aren't reliable and incomplete. You can also use the exhaustive level for the initial phase of matching and a more stringent search level for the later phase.
Typical	Use in most scenarios to search for an optimum number of match candidates.	Optimal	Appropriate for average volumes of data that are moderately critical.
Narrow	Performs a stringent search that returns fewest match candidates. This search level results in fewer match candidates than other search levels, possibly resulting in undermatching, and processes faster.	Low	Appropriate for large volumes of data that are reliable and complete or contains a large number of duplicates.

Scenarios for candidate selection criteria configuration

When you configure candidate selection criteria, consider industry, size and quality, the consequences of missing a match or overmatching, and processing time.

The following scenarios illustrate recommended settings for key generation and search level matching:

Flight reservation system

An operations manager wants to check if a passenger is on the no-fly list and a missed match can be critical. In this scenario, you can use an extended key generation and an extreme search level that results in the maximum number of keys and candidates, ensuring no potential match is missed. The process might take a while to run, but time is less important than missing a match.

Retail store

When a cashier searches for the loyalty account of a customer with other customers waiting, speed is essential but a missed match isn't critical. Use of standard key generation and a typical search level balances quick response times with adequate matching accuracy and minimal impact if a match is missed.

Financial institution

A financial institution wants to merge customer records and if they merge records incorrectly, they might compromise customer data. In this scenario, limited keys produce only a few key permutations, and a narrow search returns only extremely similar candidates. You can also avoid overmatching of records.

Guidelines for configuring candidate selection criteria

Consider the following guidelines when you configure candidate selection criteria:

- If data contains names of organizations, use the name Organization_Name field to generate match keys. If data contains names of individuals, use the Person_Name field to generate match keys. If data contains addresses, use the Address_Part1 field to generate match keys.
- If you can identify more candidates that might not be identified using a single field, use multiple fields to generate match keys.
- Use a match key width that provides you the best performance without affecting your match quality. You might have to trade off some match quality for increased performance.
- Use the narrowest possible search level to generate acceptable matches. In most cases, the typical search level might suffice.
- Ensure that you clean or suppress data that returns a large number of match candidates. You can look for large groups of identical keys when there are identical records. For example, high-frequency phone numbers, email addresses, words, or phrases can return identical keys.
- Consider filtering candidates to minimize the number of candidates for large data volumes. For example, you can filter candidates using a state or country code.

Maximum candidates

To improve the accuracy and performance of the match process, specify the maximum number of candidates that the match process can use for each record.

The default is 10,000. By default, the match process identifies up to 10,000 candidates for each record. A number higher than 10,000 might affect the performance of the match process because the match process might take time to process a large number of candidates. A number lower than 2000 can result in undermatching and might affect match accuracy.

When the number of candidates for a record exceeds the maximum limit, the match process randomly selects the candidates. The set of candidates might vary for different match and merge jobs.

Note: Effective in the May 2023 release, the ability to set the maximum number of candidates for matching is available for preview.

Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

CHAPTER 3

Configuring candidate selection criteria

Before you configure match models, create candidate selection criteria for a business entity so that you can associate them with match models. You can create candidate selection criteria separately and reuse them across match models.

You can create a maximum of 10 candidate selection criteria for each business entity. When you create a candidate selection criteria, you can specify the default population based on which the candidates are selected.

After you create a candidate selection criteria, it remains active and ready to be associated with match models. A candidate selection criteria becomes inactive only when you disable it. However, you can't disable or delete a candidate selection criteria that's associated with match models until you remove it from the models or delete the models. After you create candidate selection criteria, they appear on the Candidate Selection Criteria page.

The following table describes the properties of the candidate selection criteria that you can view on the Candidate Selection Criteria page:

Properties	Description
Default Population	Default population associated with the candidate selection criteria. The candidate selection criteria appear based on the default population. Additionally, you can expand or collapse the default population to view the match models associated within a candidate selection criteria.
Field Type	Type of field used in the candidate selection criteria.
Field Name	Name of the field used in the candidate selection criteria.
Filter Candidates By	Condition used to filter the candidates that the criteria selects.
Key Generation Level	Level of thoroughness with which the match model generates match keys to identify candidates for matching. The key generation levels are extended, limited, and standard.

Properties	Description
Number of Match Models	Number of match models that use the candidate selection criteria.
Status	Status of the candidate selection criteria. You can view one of the following statuses: - Enabled. Indicates that the candidate selection criteria is active and ready to be associated with match models. A candidate selection criteria is enabled automatically when you create it. - Disabled. Indicates that the candidate selection criteria is inactive.

Add candidate selection criteria

To select match candidates during match and merge process, add candidate selection criteria for a business entity, and then associate them with match models.

1. Open the business entity, and click the **Match** tab.
2. On the **Candidate Selection Criteria** page, click **Add Candidate Selection Criteria**.

   The **Add Candidate Selection Criteria** dialog box appears.
3. Select a default population.
4. Specify the business entity Field Name, Field Type, Filter Candidates By, Key Generation Level, and the Candidate Search Level for candidate selection.

   For more information about candidate selection criteria, see "Candidate selection criteria" on page 19.
5. Click **Add**.

Managing candidate selection criteria

After you create candidate selection criteria for a business entity, you can enable, disable, and delete them.

You can create a maximum of 10 candidate selection criteria for each business entity. You can view the list of candidate selection criteria created for a business entity on the Candidate Selection Criteria page.

You can perform one of the following actions for each candidate selection criteria listed on the page:

- Enable. Activate a candidate selection criteria so that you can associate it with match models.
- Disable. Deactivate a candidate selection criteria that you no longer need. You can enable it later if required.
- Delete. Permanently delete the candidate selection criteria that you no longer need.

You can't disable or delete a candidate selection criteria that match models use. If the criteria is a part of published match models, remove the match models before you disable or delete the criteria.

Enable candidate selection criteria

By default, candidate selection criteria remain enabled when you create them. However, if you disable a candidate selection criteria, you can enable it again and associate it with match models.

1. Open the business entity, and click the **Match** tab.
2. On the **Candidate Selection Criteria** page, select the candidate selection criteria that you want to enable.
3. Click **Actions** > **Enable**.

Disable candidate selection criteria

You can disable a candidate selection criteria that you no longer want match models to use.

1. Open the business entity, and click the **Match** tab.
2. On the **Candidate Selection Criteria** page, select the candidate selection criteria that you want to disable.
3. Click **Actions** > **Disable**.

   Note: You can't disable a candidate selection criteria that match models use. If the criteria is a part of published match models, remove the match models before you disable the criteria.

Delete candidate selection criteria

You can permanently delete a candidate selection criteria that you no longer need.

1. Open the business entity, and click the **Match** tab.
2. On the **Candidate Selection Criteria** page, select the candidate selection criteria that you want to delete.
3. Click **Actions** > **Delete**.

   Note: You can't delete a candidate selection criteria that match models use. If the criteria is a part of published match models, remove the match models before you delete the criteria.

CHAPTER 4

Configuring match models

To configure the match and merge process, start by creating a match model. Decide on the method to create a match model based on whether you use a predefined or a custom business entity.

You can use the following methods to create match models:

Method for creating match models	Business entity type
From a copy of a match model	Use for predefined business entities.
From scratch	Use for predefined and custom business entities.

Creating a match model from a copy

You can create a match model for a predefined business entity from a copy of a match model. You can create a copy of a predefined match model or a model created from scratch.

To create a match model from a copy of a match model, perform the following steps:

- "Step 1. Review the existing match model" on page 29
- "Step 2. Create a copy of a match model" on page 30
- "Step 3. Configure a match model" on page 30

Step 1. Review the existing match model

Review the predefined or a custom match model to determine whether the model meets your business requirements.

1. Open the predefined business entity, and click the **Match** tab.
2. On the **Match Models** page, click the name of the match model.

   The **Model Configuration** page appears.
3. Review the basic model configuration and the candidate selection criteria.
4. **Next step**: If the match model meets your requirements, use the model without changes. Otherwise, configure a new model starting from scratch or from a copy of the existing model.

Step 2. Create a copy of a match model

After you determine the need to customize a match model, create a copy of the model.

Note: You cannot edit the predefined match model, so create a copy of the model for customization.

1. On the **Match Models** page, click the **Copy** icon beside the name of the match model.

   The **Copy <match model name>** dialog box appears.
2. Enter a unique name for the match model, and click **Copy**.

   A copy of the model in the draft state appears on the **Match Models** page.
3. **Next step**: Configure and publish the match model.

Step 3. Configure a match model

After you create a copy of the match model, you can configure the match model.

1. From the **Match Models** page, click the copy of the match model that you want to configure.

   The **Model Configuration** page appears.
2. Configure the model name and model objective.

   Note: In the current release, **Resolve Duplicates** is the only model objective that is available.

   For more information, see "Model name" on page 12 and "Model objective" on page 12.
3. To configure the maximum number of candidates that the match process can use for each record, select **Custom** and enter a value from 1 through 1,000,000.

   For more information, see "Maximum candidates" on page 25.
4. Select a default match population to use if the records don't contain any population information.

   A detailed information about the selected population appears.
   For more information about default population, see "Default Population" on page 13.
5. Configure the candidate selection criteria.
   a. Review the candidate selection criteria in the model and edit the criteria if required.
   b. To add a criteria, choose one of the following options:
      - **Select existing**. Select an existing criteria that uses the default match population you selected in step 4.
      - **Create new**. Create a new candidate selection criteria based on the default match population.
   c. If you choose **Select existing**, click **Select** in the **Select Criteria** field.
   d. In the **Select Criteria** dialog box, select the candidate selection criteria that you want to use, and click **Select**.

e. If you choose **Create new**, specify the business entity Field Name, Field Type, Filter Candidates By, Key Generation Level, and the Candidate Search Level for candidate selection.

You can't select a field that's configured as a calculated attribute. For more information about fields that are configured as calculated attributes, see *Define the Data Model*.

Note: Effective in the November 2023 release, the ability to configure fields as calculated attributes is available for preview.

Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.

For more information about candidate selection criteria and how to create them, see "Candidate selection criteria" on page 19 and Chapter 3, "Configuring candidate selection criteria" on page 26.

6. Click **Add**.
7. **Next step**: Review and configure the Directed AI match rules.

Creating a match model from scratch

You can create a match model from scratch for both custom and predefined business entities.

1. Open the business entity, and click the **Match** tab.
2. On the **Match Models** page, click the **Add Match Model** icon.

   The **Add Match Model** wizard appears.
3. Configure the model name and model objective.

   Note: In the current release, **Resolve Duplicates** is the only model objective that is available.

   For more information, see "Model name" on page 12 and "Model objective" on page 12.
4. To configure the maximum number of candidates that the match process can use for each record, select **Custom**, and enter a value from 1 through 1,000,000.

   For more information, see "Maximum candidates" on page 25.

   Note: Effective in the May 2023 release, the ability to set the maximum candidates for matching is available for preview. Preview functionality is supported for evaluation purposes but is unwarranted and is not supported in production environments or any environment that you plan to push to production. Informatica intends to include the preview functionality in an upcoming release for production use, but might choose not to in accordance with changing market or technical circumstances. For more information, contact Informatica Global Customer Support.
5. Select a default match population to use if the records don't contain any population information.

   A detailed information about the selected population appears.
   For more information about default population, see "Default Population" on page 13.
6. Click **Next**.

7. On the **Add Candidate Selection Criteria** page, configure candidate selection criteria.
   a. To add a criteria, choose one of the following options:
      - **Select existing**. Select an existing criteria that uses the default match population you selected in step 5.
      - **Create new**. Create a new candidate selection criteria based on the default match population.
   b. If you choose **Select existing**, click **Select** in the **Select Criteria** field.
   c. In the **Select Criteria** dialog box, select the candidate selection criteria that you want to use, and click **Select**.
   d. If you choose **Create new**, specify the business entity Field Name, Field Type, Filter Candidates By, Key Generation Level, and the Candidate Search Level for candidate selection.

      For more information about candidate selection criteria and how to create them, see "Candidate selection criteria" on page 19 and Chapter 3, "Configuring candidate selection criteria" on page 26.
8. To open the match model immediately after the model is created, select **Immediately open the model**.
9. Click **Add**.

   If you selected **Immediately open the model**, the match model opens. Otherwise, the match model appears on the **Match Models** page.
10. **Next step**: Configure the Directed AI match rules and train the Adaptive AI match model.

Editing a match model

You can modify match models that are in draft status. When you edit a match model, you create another version of the model.

You can have a maximum of 50 versions for a match model. If the number of match model versions reaches the maximum limit, delete an existing version and create another version, or copy the model to create a new model. To change the maximum limit of model versions, contact Informatica Global Customer Support.

1. Open the business entity, and click the **Match** tab.

   The list of match models appears.
2. Click the match model that you want to edit, and click **Edit**.
3. In the **Model Configuration** section, edit the properties.
4. In the **Candidate Selection Criteria** section, hover over the row that you want to edit, and click **Edit**.
5. Edit the required criteria, and click **Update**.
6. To add candidate selection criteria, click **Add Criteria**.
7. Click **Save**.

Deleting a match model

You can delete the match models that you don't need. Your user role must have the required privileges to delete the match models. If a match model has multiple versions, you can delete a specific version without impacting the other versions. You can't delete the predefined match models.

Assets such as match and merge jobs and Search Match APIs might use a published match model that you want to delete. Before you delete the match model, remove the match model from these dependent assets.

1. Open the business entity for which you want to delete a match model, and click the **Match** tab.

   The list of match models appears.
2. Hover over the match model that you want to delete, and click **Delete**.
3. If the match model has dependent assets, remove the dependencies and repeat step 2.

   To remove dependencies, you can perform the following tasks for the dependent assets:

Dependent Asset Type	Task
Job	Use a different version of the match model or another match model in the job. **Note:** If you delete the dependent job, you can't delete the match model.
Search Match API	Use a different version of the match model or another match model as the default model for Search Match API. **Note:** Business 360 Console can't detect external Search Match APIs that are dependent on the match model that you want to delete. Ensure that you remove dependencies for these APIs as well.

4. Click **Delete**.

   The match model is deleted from the list.

Distinct source systems

You can configure a source system as distinct so that records from the source system don't match each other but match records from other source systems. Distinct source systems help reduce unwanted matching of records.

If most of the records in a source system are unique, you can configure the source system as distinct to avoid matching of records within the source system. For example, the Dun & Bradstreet database provides unique data. To avoid matching of these records, you can configure the Dun & Bradstreet database as distinct.

However, you can allow records from a distinct source system to match each other by the Directed AI match rules that identify record pairs for manual merge. The users can manually review these record pairs.

For example, consider a match model with two Directed AI match rules, Rule 1 and Rule 2. The merge strategy of Rule 1 is set to manual, and the merge strategy of Rule 2 is set to automated. Set the HR source system as distinct. In this case, Rule 1 and Rule 2 don't match the records that belong to the HR source system.

If you allow records from the HR source system to match each other by the Directed AI match rules that identify record pairs for manual merge, Rule 1 matches records that belong to the HR source system. The identified record pairs are sent for manual review.

Configuring a distinct source system

You can configure a source system as distinct so that the records that belong to this source system don't match each other.

1. Open a business entity, and click the **Match** tab.

   The list of match models appears.

2. Click **Source System Settings**.

   The **Source System Settings** page appears with a list of source systems.

3. Select **Distinct for All Merge Strategies** for the source systems that you want to configure as distinct.

   The source systems become distinct and doesn't match the records with any Directed AI match rules.

4. To allow records from a distinct source system to match each other by the Directed AI match rules that identify record pairs for manual merge, select **Exclude Manual Merge Strategy**.

5. Click **Save**.

CHAPTER 5

Directed AI match rule properties

Directed AI match rules are a set of conditions and business entity fields that are required to identify duplicate records.

You can configure the following Directed AI match rule properties:

- "Description" on page 35
- "Match strategy" on page 36
- "Merge strategy" on page 36
- "Match criteria" on page 37
- "Match level" on page 39
- "Merge threshold" on page 37

Additionally, you need to configure the business entity fields based on which you want to match records. These business entity fields are called the match fields. For more information, see "Match fields" on page 41.

Description

A succinct description helps identify a Directed AI match rule. The description can include the purpose of the rule.

Description is an optional Directed AI match rule property but recommended for differentiating multiple rules.

Match strategy

A match strategy indicates the type of match you want to perform. You can decide on an exact or a fuzzy match strategy based on the quality and characteristics of your data.

The following table describes the match strategies:

Match Strategy	Description
Exact	A deterministic match that includes records containing identical values. Configure the exact match strategy if the quality of the data is good. For example, use the exact match strategy if at least some fields contain unique data, such as IDs, postal codes, and industry codes.
Fuzzy	A probabilistic match that includes variations in data patterns, such as spelling variations, misspellings, transpositions, omissions, and phonetic variations. To ensure that the match process does not skip record pairs with data variations, but are a match, configure this match strategy. For example, use the fuzzy match strategy to match records based on fields that contain addresses, geographic coordinates, and names of people or organizations.

Merge strategy

A merge strategy indicates the action to be performed on the records that have gone through the match process. The process uses the merge thresholds, which are match score ranges, to determine the merge strategy. Based on the merge strategy and the associated merge threshold of the rule, the record pairs result in automated, manual, threshold-based, or skipped merge.

The following table describes the merge strategies:

Merge Strategy	Description
Automated	Automatically merges the records identified as duplicates if their match scores fall within the specified merge threshold range for automated merge. For example, use the automated merge strategy for Directed AI match rules that are based on unique identifier fields, such as social security number. Two people cannot have the same Social Security number, so you do not have to manually review the records.
Manual	Queues the records identified as duplicates for manual merge if their match scores fall within the specified merge threshold range for manual merge. Users with the required privileges can review and merge the records in the Business 360 applications. For example, use the manual merge strategy for Business 360 applications that are based on fields, such as the address field. Multiple family members can share the same address and the records cannot be definite matches. You need to manually review the records to confirm whether these are duplicate records.

Merge Strategy	Description
Threshold-based	Available only when you select the fuzzy match strategy. Based on the match score of the record pairs, the match and merge process determines whether the result is automated, manual, or skipped merge. For example, use the threshold-based merge strategy for a Directed AI match rule that is based on a field, such as address, that is quite similar but other field values are not. In this scenario, you can set the thresholds to skip merging records when the match score is 70 and lower, automate merging records when the match score is 90 and higher, and send for manual merge when the match score is from 71 to 89.
Skip	The records in the pairs are not matched against each other if their match scores fall within the range set for the merge threshold for skipping merge.

Merge threshold

Applicable to Directed AI match rules with the fuzzy match strategy. Merge thresholds are based on match score ranges of records that undergo matching. You configure merge thresholds for the merge strategy that you select for a Directed AI match rule. When the system identifies records that meet the corresponding merge threshold, the merge strategy that you configure is applied. A higher merge threshold value indicates a higher strength of the merge strategy that you choose.

The following table describes the merge thresholds:

Merge Threshold	Description
Automated Merge	Determines whether the record pairs can merge automatically. Set the range with the lower limit set to a high match score. To ensure high accuracy, set a smaller range. When the match scores fall within the range set for automated merge, the matching record pairs merge automatically.
Manual Merge	Determines whether the record pairs can be queued for manual merge. Set a match score range for identifying record pairs that need a manual review before being merged. The range can be wider than the range for automated merge. When the match scores fall within the range set for manual merge, the matching record pairs are queued for manual merge.
Skip Merge	Determines whether the record pairs can be skipped from merging and are not matched against each other. Set a match score range for identifying record pairs that need to be skipped. When the match scores fall within the range set for skipping merge, the records in the pairs are not matched against each other.

Match criteria

Applicable to Directed AI match rules with a fuzzy match strategy. A match criteria is the determining factor behind matching records for your business needs. The types of fields that are required for matching are

based on the match criteria that you select. For every fuzzy rule that you define, you must select a criteria from a list of predefined match criteria.

For example, when address is important to determine whether two records are for the same person, use the Resident match criteria. The system compares records based on the Resident match criteria to determine whether the records are a match.

Two Directed AI match rules that have the same properties except for the match criteria result in different sets of matching records.

Match criteria vary based on the match population of your input data.

The following table describes the match criteria for the usa match population:

Match Criteria	Description
Address	Identifies an address match. The address might be postal, residential, delivery, descriptive, formal, or informal.
AuthorISBN	Matches an author for the specified book with an International Standard Book Number (ISBN).
CC_Issuer	Matches the organization that issues credit cards to the customers. Use this match criteria to match credit cards or the credit card-issuing organizations.
CC_Owner	Matches the credit card owner and the address.
Contact	Matches a contact within an organization at a specific location.
Corp_Entity	Matches an organization based on its legal corporate name, including the legal endings such as INC and LTD. Use the Corp_Entity match criteria to honor differences between names, such as ABC TRADING INC and ABC TRADING LTD.
Division	Matches an organization based on the address. This match criteria matches organization X at address Y or Z if multiple addresses are provided.
Family	Matches individuals with the same or similar family names, who share the same address or the same telephone number.
Fields	For general nonspecific use. Use this criteria with any field type.
Geocode	Matches records based on geocode, which is the geographic coordinates and elevation.
Household	Matches individuals with the same or similar family names, who share the same address.
Individual	Matches individuals by name and with either the same ID or date of birth.
Organization	Matches organizations primarily by name. Use this match criteria for online searches where a human is available to review the search results and make a choice.
Person_Name	Matches a person by name. Use this match criteria for online searches where a human is available to review the search results and make a choice.
PublisherISBN	Matches the publishing organization for a specified book with an International Standard Book Number (ISBN). The ISBN can be a 10-digit number or a 13-digit number.
Relationship	Matches records based on the relationship between business entities.
Resident	Matches a person based on an address.

Match Criteria	Description
VIN_Manufacturer	Matches the manufacturer of a vehicle for a specified vehicle number.
VIN_Owner	Identifies and matches the owner of a vehicle.
Wide_Contact	Identifies a contact within an organization without regard to the location.
Wide_Household	Identifies individuals who share the same address, the same family name, or the same phone number.

Match level

Applicable to Directed AI match rules with a fuzzy match strategy. The match level indicates the level of fuzzy matching that you want to perform on your data. The number of matching records that a Directed AI match rule generates depends on the match level.

The following table describes the match levels:

Match Level	Description
Conservative	Performs stringent matching that results in few matching records. Some matching records might slip through the match process.
Loose	Performs lenient matching that results in a significant number of matching records. Some records might be incorrect matches. Use the loose match level in a Directed AI match rule with a manual merge strategy to capture the potential matching records that the stringent match rules might miss. A user can manually review these matching records before merging them.
Typical	Use in most matching scenarios to get an optimal number of matching records.

Rank

After you configure all the Directed AI match rules, rank them in the order in which you want to run them for matching record pairs. Based on the ranking, all the Directed AI match rules complete processing one record pair before processing the next record pair. The Directed AI match rules are run on one record pair at a time.

First, the Directed AI match rule with rank 1 is used to process a record pair. If the record pair meets the conditions set by the match rule, the merge strategy for the rule is applied to the record pair. The processing for the record pair is complete.

If the record pair does not meet the conditions set by the match rule, then the rule that is next in rank is used to process the record pair.

If the record pair does not meet the conditions set by any of the ranked rules, the unranked rules, which are the rules with the threshold-based merge strategy, are applied to the record pair. All the unranked rules are applied in succession, and the outcome is assessed to make a unanimous decision.

The following table lists the actions on record pairs for the unranked, threshold-based rule outcomes:

Unranked Rules Outcome	Action on Record Pair
Majority of the rules have the same outcome	Majority outcome is applied. For example, if the majority outcome is automated merge, then the record pair is queued for automated merge.
Tie between manual merge and automated merge	Queued for manual merge.
Tie between automated merge and skipped merge	Queued for manual merge.
Tie between manual merge and skipped merge	Queued for manual merge.
Tie between manual merge, automated merge, and skipped merge	Queued for manual merge.

For example, you have a set of threshold-based match rules, and you configure the following match score ranges as the merge threshold:

- Automated Merge. Match score between 96 and100.
- Manual merge. Match score between 70 and 95.
- Skipped Merge. Match score less than 70.

The following table lists the actions on record pairs when there is a tie between the threshold-based rule outcomes:

Match Scores	Unranked Rules Outcome	Action on Record Pair	Match Report
Manual Merge Rule 1 - 93 Manual Merge Rule 2 - 95 Automated Merge Rule 3 - 97 Automated Merge Rule 4 - 99	Tie between the number of manual merge and automated merge outcomes	Queued for manual merge	Manual Merge Rule 2 Match Score - 95 (Highest match score among the manual merge outcomes.)
Automated Merge Rule 1 - 97 Automated Merge Rule 2 - 99 Skipped Merge Rule 3 - 47 Skipped Merge Rule - 49	Tie between the number of automated merge and skipped merge outcomes.	Queued for manual merge (Since the system is unable to decide the outcome.)	No match rule Match Score - NA

Match Scores	Unranked Rules Outcome	Action on Record Pair	Match Report
Manual Merge Rule 1 - 93 Manual Merge Rule 2 - 95 Skipped Merge Rule 3 - 47 Skipped Merge Rule 4 - 49	Tie between the number of manual merge and skipped merge outcomes.	Queued for manual merge	Manual Merge Rule 2 Match Score - 95 (Highest match score among the manual merge outcomes.)
Manual Merge Rule 1 - 93 Automated Merge Rule 2 - 97 Skipped Merge Rule 3 - 49	Tie between the number of manual merge, automated merge, and skipped merge outcomes.	Queued for manual merge	Manual Merge Rule 1 Match Score - 93 (If you have multiple match rules with the manual merge outcome, the match rule with the highest match score is selected.)

Match fields

Match fields are business entity fields that you use in Directed AI match rules to identify matching records. You can configure one or more match fields for each Directed AI match rule.

The following table describes the types of match fields that you can configure:

Match Field	Description
Exact	Business entity fields that contain unique data, such as IDs, postal codes, and industry codes. Use exact match fields for matching identical data. You can configure exact match fields when you chose an exact or fuzzy match strategy.
Fuzzy	Business entity fields that contain data with variations, such as addresses, geographic coordinates, and names of people or organizations. Use fuzzy match fields for identifying data that are similar. You can configure fuzzy match fields only when you chose the fuzzy match strategy.

Note: When you configure the match and merge process, avoid using multiline fields and text fields that contain rich text format as match fields. Such fields can contain longer values that might affect match scores.

The following table describes the match field properties:

Property	Description
Field Name	Name of the business entity field that you want to use for identifying duplicate records.
Field Type	Applicable to Directed AI match rules with a fuzzy match strategy. The field type indicates the type of data a field contains. When you configure the match criteria for a fuzzy Directed AI match rule, the required field types are automatically added for the fuzzy match fields.
Comparison Outcome	The result of comparing records to determine whether they are a match. When comparing records with fuzzy match fields, the records that are identified as matches are similar. When comparing records with exact match fields, the records that are identified as matches are identical.

Field type

A field type indicates the type of data a business entity field contains. The fuzzy matching process is effective only when you configure appropriate field types for the match fields.

Each match criteria requires some field types. The following table describes the field types:

Field Type	Description
Address_Part1	Contains address details, such as building name, street number, street name, street type, and apartment details.
Address_Part2	Contains address details, such as city or town name, state name, ZIP code, and country.
Attribute_1|Attribute_2	Contains generic data.
Code	Contains numeric or alphanumeric values, such as license number or vehicle registration number.
CreditCard	Contains credit card numbers.
Date	Contains dates, such as date of birth, expiry date, date of contract, date of change, or issued date.
Email	Contains email addresses.
Geocode	Contains geographic coordinates, such as latitude, longitude, and elevation. For more information about the Geocode field type, see "Geocode field type" on page 21.
ID	Contains IDs, such as account numbers, registration numbers, license numbers, policy numbers, or Social Security numbers.
ISBN10	Contains the 10-digit International Standard Book Number (ISBN).
ISBN13	Contains the 13-digit ISBN.
Model_Number	Contains model numbers of products.
Organization_Name|Company_Name	Contains organization names, such as business names, institution names, department names, agency names, or trading names.

Field Type	Description
Person_Name	Contains names of individuals.
Postal_Area	Contains ZIP codes of towns or cities.
Product_Description	Contains description of products.
Product_Name	Contains names of products.
Telephone_Number	Contains telephone numbers.
VIN	Contains Vehicle Identification Numbers (VINs).

Additional required fields and optional fields

Applicable to Directed AI match rules with a fuzzy match strategy. When you add a Directed AI match rule, you must configure the additional required fields along with the required fields.

Additional required fields are required fields but requires at least one of the fields to be configured. For example, for the **CC_Issuer** criteria, **CreditCard** is a required field, and **Company_Name** and **Organization_Name** are additional required fields. You must select at least one of these additional required fields and map the field name.

You can also add optional fields. For example, for the **Wide_Contact** criteria, you can add the following additional fields:

- Code
- Email
- Id
- Attribute1
- Attribute2

The additional required fields and optional fields appear based on the match criteria that you select.

Field properties

You can configure the properties of the match fields to enhance match accuracy. You can enable segment matching or null matching on exact match fields. Segment matching and null matching are mutually exclusive. You can configure radius for Geocode fields to determine the proximity of the addresses.

Segment matching

You can configure a Directed AI match rule to limit matching to specific subsets or segments of data.

You can segment data based on the values of exact match fields, such as country and gender, for the match process to identify matching records within these segments. The match process ensures matching records within the segment specified in the Directed AI match rule. For example, you can segment data by country to

create a subset for Japan. The match process then compares records within the Japan segment to identify matching records.

You can also define multiple segments based on multiple distinct values of an exact match field. The match process then matches records within each segment and across segments.

For example, if you enable segment matching on the Country field and define segments for Japan and France, the match process matches records in the following ways:

- Matches records within the Japan segment.
- Matches records within the France segment.
- Matches records between the Japan and France segments.

However, if you don't want to match records between the Japan and France segments, create separate Directed AI match rules for each segment.

You can also match the data of a segment with the rest of the data. When you enable a Directed AI match rule to match segment data with the rest of the data, the match process performs the following tasks:

- Matches the records within the segment.
- Matches the records of the segment with records that don't belong to the segment.

For example, you can match records that contain Japanese addresses with records that contain addresses from other countries, such as France and Germany.

Note: In Directed AI match rules, if you enable segment matching for picklist fields, such as Country and City, ensure that the values aren't deleted from the associated code lists. Otherwise, the job fails.

Segment matching scenario

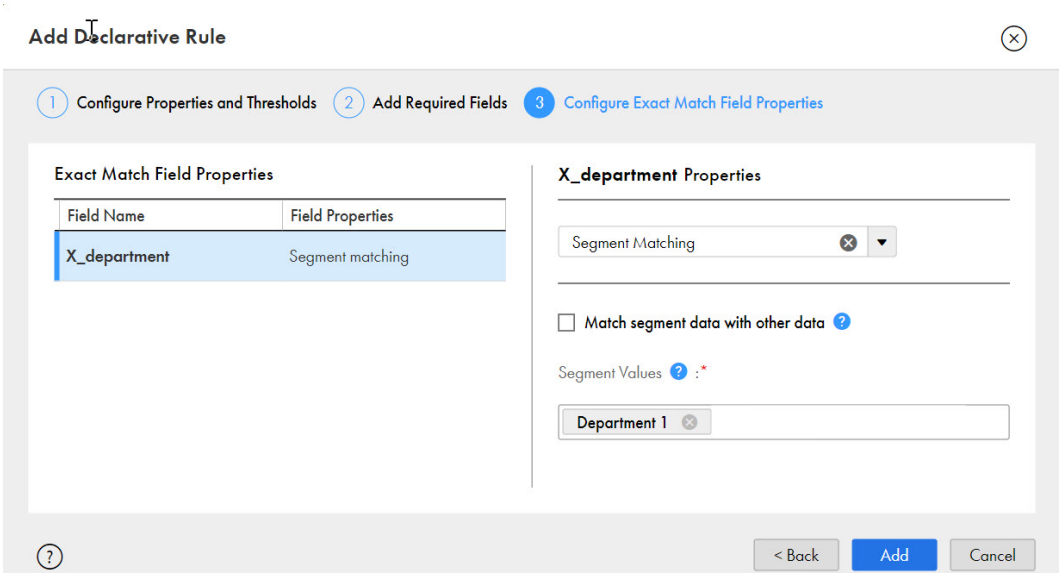
A custom business entity in your organization contains records of employees that belong to different departments. You want to match employee records based on their departments.

The following table shows the employee records and the departments they belong to:

Record	Employee Name	Department
Record 1	John Lee	Department 1
Record 2	John Lee	Department 1
Record 3	Tom Jones	Department 1
Record 4	Tim Arnold	Department 2
Record 5	John Lee	Department 2
Record 6	John Lee	Department 2
Record 7	Tom Rob	Department 3
Record 8	John Lee	Department 3
Record 9	John Lee	Department 3
Record 10	Tom Jones	Department 1

When you enable segment matching for the Department field and specify Department 1 as the segment value within a Directed AI match rule, the match process matches records of employees that belong to Department 1.

The following image shows the Add Directed AI Match Rule dialog box with segment matching enabled for the Department 1 field within a Directed AI match rule:

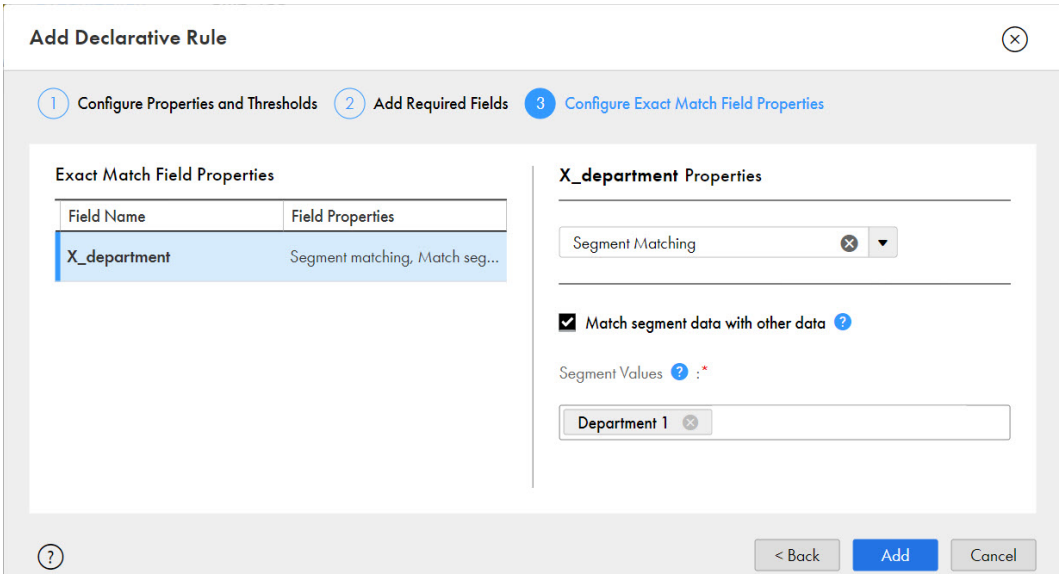


The following table lists the match pairs generated based on the Directed AI match rule:

Match Pair	Records
Match pair 1	Record 1 and Record 2
Match pair 2	Record 3 and Record 10

When you enable segment matching to match the data of a segment with other data, the match process matches the employees of Department 1 with the employees of other departments.

The following image shows the Add Directed AI Match Rule dialog box with segment matching enabled for the Department 1 field to match employee records of other departments:



The following table lists the match pairs generated based on the Directed AI match rule when it's enabled to match segment data with the rest of the data:

Match Pair	Records
Match pair 1	Record 1 and Record 2
Match pair 2	Record 1 and Record 5
Match pair 3	Record 1 and Record 6
Match pair 4	Record 1 and Record 8
Match pair 5	Record 1 and Record 9
Match pair 6	Record 2 and Record 5
Match pair 7	Record 2 and Record 6
Match pair 8	Record 2 and Record 8
Match pair 9	Record 2 and Record 9
Match pair 10	Record 3 and Record 10

In the preceding example, records 5 and 6 and 8 and 9 aren't matched because the department of the employees isn't Department 1.

Additional settings for segment matching

When you enable segment matching for multiple fields, you can specify whether you want to segment data based on all match fields or any of the match fields.

To segment data when multiple match fields are enabled for segment matching, you can select one of the following conditional operators:

- Match All Fields (AND Operation). Segments data based on all segment matching-enabled match fields. For example, if you enable three match fields, Country, State, and Marital Status, for segment matching, the segment data contains records that contain all the specified segment values.
- Match Any Field (OR Operation). Segments data based on any of the segment matching-enabled match fields. For example, if you enable three fields, Country, State, and Marital Status, for segment matching, the segment data contains records that contain at least one of the specified segment values.

Null matching

Use null matching to enable the exact match fields that have no values to participate in the match process. You can configure null values to match other null values or non-null values. If you don't use null matching, the fields with null values don't participate in the match process.

Note: When you configure multiple Directed AI match rules with null matching enabled in a match model, the match process retains only one null match pair for each source record. This behavior prevents the formation of inaccurate or large clusters.

You can configure the following types of null matching:

- Null matches null. The match process matches two null values.
  You can select one of the following options for the match process to retain match pairs that contain null values:
  - Match pair with the highest score. Retains the null value match pair with the highest score in a record pair group.
    When you configure a Directed AI match rule to retain match pairs with the highest score, the match process retains the pair with the highest score and deletes the rest. When you run a match and merge job again, the records in the deleted pairs don't participate in the match process because their consolidated status is set to matched. To enable their participation in the match process, you need to reset their status.

    Also, the match process marks all records that participated in matching with the name of the rule and match model that they matched with. For example, consider a match model named MODEL_A that includes a Directed AI match rule called RULE_A, configured to retain the match pair with the highest score. If you run a match and merge job using MODEL_A, then the match process marks all records that participated in matching as RECORD1_MODEL_A_RULE_A, RECORD2_MODEL_A_RULE_A, and so forth.

    If you match the same records again with MODEL_A, the match process skips RULE_A and matches them with other rules in the match model to prevent overmatching. The match process functions the same way even when you create newer versions of MODEL_A or modify the null matching configuration in RULE_A.

    If you want to match the records with the same rule again, you can create a new match model and add the Directed AI match rule to that model.
  - All match pairs. Retains all the null value pairs in a record pair group, irrespective of their score. This option reduces the chances of forming individual master records from null value match pairs with lower scores. To avoid overmatching, ensure that you add fuzzy fields in addition to exact match fields.

    Note: To optimize match results and avoid potential undermatching, choose to retain all match pairs.
- Null matches non-null. The match process matches a null value with a non-null value.
  When a record with null values matches with multiple records with non-null values, the match process considers the record pair that has the highest match score. If multiple record pairs have the same match score, the match process considers the last updated record to form a match pair.

Null matching scenario

You have a list of client records, and some of the fields don't contain any data. In a Directed AI match rule, you select Person Name as a fuzzy match field and Phone Number, Address, and SSN as exact match fields. You enable null matching for the SSN field.

The following table shows the sample records that you want to match:

Record	Person Name	Phone	Address	SSN
1	James A Adam	4196251490	123 main st	-
2	J Adam	4196251490	123 main st	-
3	James Adam	4196251490	123 main st	-
4	James Alan Adam	4196251490	123 main st	-

When you configure null matches null, the match process considers the following records as match pairs:

- Records 1 and 2
- Records 3 and 4

If you choose to retain the match pair with the highest score, it might cause undermatching because the match process retains only one match pair for each source record and deletes the rest. The merge process merges records 1 and 2 into one master record and records 3 and 4 into another master record.

To reduce this risk of undermatching, choose to retain all match pairs so that the match process retains all the match pairs. The merge process then merges records 1, 2, 3, and 4 into a single master record.

You have a list of client records, and some of the fields don't contain any data. In a Directed AI match rule, you select Person Name as a fuzzy match field and Phone Number and Extension as exact match fields. You enable null matching for the Extension field.

The following table shows the sample records that you want to match:

Record	Person Name	Phone	Extension
1	John Smith	6053128215	-
2	John Smith	6053128215	-
3	John Smith	6053128215	254
4	Jon Smiths	6053128215	254
5	Joanne Soul	6053128216	256

When you configure null matches non-null, the match process considers the following records as match pairs:

- Records 1 and 3
- Records 2 and 3
- Records 3 and 4

The person name values for records 1 and 4 don't match exactly, so the match score is less as compared to records 1 and 3. The match process considers records 1 and 3 as match pair.

Note: If a null value matches another null or a non-null value, you might be unsure if the two records are a match. To verify that records are a match, use manual merge strategy for Directed AI match rules with null matching enabled.

Radius

You can configure a Directed AI match rule that uses the Geocode fields to match addresses based on their geographic coordinates.

You can specify a radius to determine the proximity of addresses. The value can be between 1 to 10,000 meters. Default is 1000.

Guidelines for configuring Directed AI match rules

Consider the following guidelines when you configure Directed AI match rules:

- Analyze your data and determine whether you need an automated or a manual merge strategy.
- Before configuring match rules, profile the data properly. Optimal match rules are derived after iterative tests.
- Create simple Directed AI match rules before moving to complex ones. With this approach, you can refine and improve your match strategy based on the results.
- Use multiple Directed AI match rules to cater to different scenarios and levels of accuracy. For example, you can create a rule for exact matches, another for fuzzy matches, and additional rules for specific attributes or patterns. You can combine these rules using logical operators, such as AND and OR to define the overall match strategy.
- Add one or more exact match fields, such as postal codes or phone numbers to act as filters for each rule. Exact match fields can significantly improve performance by reducing the number of fuzzy matches.
- Use the fewest possible number of Directed AI match rules in a match model for optimal match results.

CHAPTER 6

Configuring Directed AI match rules

You configure Directed AI match rules to determine records that are duplicates and to merge them.

You can define a Directed AI match rule for exact or fuzzy matching. An exact rule matches records that have identical values in match fields. A fuzzy rule matches similar records based on probabilistic match determinations that consider likely variations in data patterns, such as misspellings, transpositions, omissions, and phonetic variations. The rules that you configure depend on the characteristics of the data, and the specific match and merge requirements.

To configure Directed AI match rules, perform the following tasks:

- "Step 1. Review the data" on page 50
- "Step 2. Add Directed AI match rules" on page 51
- "Step 3. Set the execution order for Directed AI match rules" on page 54

Step 1. Review the data

Before you add the Directed AI match rules, review and understand your data. Your data might be incorrect, inconsistent, and have missing values. Review the data for quality of the data values, consistency, uniqueness, and logic.

For example, to add Directed AI match rules to match individuals, you must understand the attributes related to individuals. If the First Name and the Last Name fields have missing values, it is not ideal to base the rules on these fields because of missing values. The Full Name field might have values for all records and is good to use as a match field in the rules.

You can use the Address fields, such as Address Line 1, City, and Postal Code, as match fields in the rules. The Address fields would help identify duplicate records. Based on your review, you can add rules to match individuals based on the name and address fields.

Step 2. Add Directed AI match rules

To match and merge duplicate records, add Directed AI match rules to the match model. For example, to match duplicate individuals based on name and address, add a fuzzy rule with the Resident match criteria.

Before you add Directed AI match rules to a match model, review any rules that the match model might include for the business entity. You might want to edit or delete any existing rules.

You can't configure Directed AI match rules for dynamic fields.

To create a Directed AI match rule, perform the following tasks:

1. "Step 1. Configure match properties and threshold" on page 51
2. "Step 2. Add required fields" on page 52
3. "Step 3. Add optional fields" on page 53
4. "Step 4. Configure field properties" on page 53

Step 1. Configure match properties and threshold

When you configure Directed AI match rules, ensure that you select a suitable match strategy. If you select a fuzzy match strategy, ensure that you select the correct match criteria as well. After you save the rules, you can't change the match strategy and the match criteria.

1. Open the business entity, and click the **Match** tab.

   The **Match Models** page appears.
2. Click the name of the match model to which you want to add the Directed AI match rules.
3. On the **Directed AI Match Rules** page, click **Edit**.
4. Click the **Add Directed AI Match Rule** icon.

   The **Add Directed AI Match Rule** wizard appears.
5. On the **Configure Properties and Thresholds** page, select one of the following match strategies:

Option	Description
Exact	A deterministic match that includes records containing identical values.
Fuzzy	A probabilistic match that includes variations in data patterns, such as misspellings, omissions, and phonetic variations.

6. If you select **Fuzzy**, skip to step 8.
7. If you select the exact match strategy, set the following properties for the Directed AI match rule:
   - "Merge strategy" on page 36
   - "Description" on page 35
8. If you select the fuzzy match strategy, set the following properties for the rule:
   - "Merge strategy" on page 36
   - "Match criteria" on page 37
   - "Match level" on page 39
   - "Description" on page 35
   - "Merge threshold" on page 37

9. Click **Next**.

   The **Add Required Fields** page appears.

Step 2. Add required fields

Add the required fields based on the match strategy that you selected. If you selected an exact or fuzzy match strategy, you can configure exact match fields. If you selected the fuzzy match strategy, you can configure fuzzy match fields only.

If the match field that you add belongs to a field group that has fields set as deduplication criteria, Business 360 Console recommends adding the deduplication criteria fields as exact match fields. If you delete an exact match field that triggered the recommendations, Business 360 Console recommends deleting previously recommended fields from the exact match fields list. If you delete a deduplication criteria field of a field group, Business 360 Console might recommend deleting the previously recommended fields from the exact match fields list.

For example, the Address field group contains Address Line 1, Postal Code, and Country Code fields, and the Postal Code and Country Code fields are configured as deduplication criteria. If you add the Address Line 1 field as a match field, Business 360 Console recommends adding the Postal Code and Country Code fields as exact match fields.

If you delete the Address Line 1 field from the Directed AI match rule, Business 360 Console recommends deleting the Postal Code and Country Code fields from the exact match fields list.

1. Click the **Add Exact Match Field** icon.

   A row is added to the **Exact Match Fields** table with the **Comparison Outcome** value set to **Identical**.

   Note: In the current release, **Identical** is the only comparison outcome that is available for exact match fields.

   For more information, see "Match fields" on page 41.

2. Select a business entity field for matching identical data.

   The field appears in the **Exact Match Fields** table. If the field belongs to a field group that has fields set as deduplication criteria, recommendations to add the deduplication criteria fields as exact match fields appear.

3. If you want to add the recommended fields as exact match fields, click the recommended fields.

   The fields appear in the **Exact Match Fields** table.

4. To delete an exact match field, hover over a row and click **Delete**.

   If you delete an exact match field that triggered the recommendations, a recommendation to delete the associated exact match fields appears.

5. To add more exact match fields, repeat steps 1 through 2.

6. If you set the Directed AI match rule to perform exact matching, click **Next**.

   The **Configure Exact Match Field Properties** page appears.

7. If you set the Directed AI match rule to perform fuzzy matching, perform the following steps:

   a. In the **Fuzzy Match - Required Fields** table, select a field name for each field type for matching similar data.

      The field appears in the **Fuzzy Match - Required Fields** table. If the field belongs to a field group that has fields set as deduplication criteria, recommendations to add the deduplication criteria fields as exact match fields appear.

   b. If you want to add the recommended fields as exact match fields, click the recommended fields.

The fields appear in the **Exact Match Fields** section.

c. In the **Fuzzy Match - Additional Required Fields** table rows, click the **Add Additional Required Field** icon to add at least one of the additional required fields.

The additional required fields appear based on the match criteria that you select.

d. Select a field for the corresponding additional required field.

A row is added to the **Fuzzy Match - Additional Required Fields** table with the **Comparison Outcome** value set to **Identical**. If the field belongs to a field group that has fields set as deduplication criteria, recommendations to add the deduplication criteria fields as exact match fields appear.

Note: In the current release, **Identical** is the only comparison outcome that is available for fuzzy match fields.

For more information, see "Match fields" on page 41.

e. Click the recommended fields if you want to add them as exact match fields.

The fields appear in the **Exact Match Fields** table.

f. To delete an additional required field, hover over a row and click **Delete**.

If you delete an additional required field that triggered the recommendations, a recommendation to delete the associated exact match fields appears.

g. Click **Next**.

The **Add Optional Fields** page appears.

Step 3. Add optional fields

If you set the Directed AI match rule to perform fuzzy matching, you can add optional fields and configure comparison outcome for them.

1. Click the **Add Optional Field** icon and select an optional field type.

   The field types appear based on the match criteria that you select.

2. Select a business entity field for the selected field type.

   The field appears in the **Fuzzy Match - Optional Fields** table. If the field belongs to a field group that has fields set as deduplication criteria, recommendations to add the deduplication criteria fields as exact match fields appear.

3. Click the recommended fields if you want to add them as exact match fields.

   The fields appear in the **Exact Match Fields** table of the **Add Required Fields** tab.

4. To delete an exact match field, hover over a row and click **Delete**.

   If you delete an optional field that triggered the recommendations, a recommendation to delete the associated exact match fields appears.

5. To add more optional fields, repeat steps 1 through 2.

6. Click **Next**.

   The **Configure Exact Match Field Properties** page appears.

Step 4. Configure field properties

You can configure the properties of match fields that you add to a Directed AI match rule. You can enable segment matching or null matching on exact match fields.

1. Select a match field for which you want to configure properties.

2. To configure segment matching, perform the following steps:
   a. On the **<field name> Properties** section, select **Segment matching**.
   b. To match segment data with the rest of the data, select **Match segment data with other data**.

      If you select **Match segment data with other data**, the rules also match the segment data with the rest of the data, but the records in the rest of the data don't match with each other. If you do not select **Match segment data with other data**, the rules use only the records in the subset of data for matching.
   c. Enter one or more segment values to identify segment data for matching.
3. To configure null matching, perform the following steps:
   a. On the **<field name> Properties** section, select **Null matching**.
   b. Select one of the following options:
      - Null matches null. Matches two null values.
      - Null matches non-null. Matches a null value with a value that is not null.
4. Click **Add**.

   The rule appears on the **Directed AI Match Rules** page.
5. Click **Save**.

Step 3. Set the execution order for Directed AI match rules

After you configure all the Directed AI match rules, set the execution order for the rules. The execution order is the rank you assign to a rule. You cannot assign ranks to Directed AI match rules that have the threshold-based merge strategy, and these rules are executed after all the ranked rules are executed.

1. Open the business entity, and click the **Match** tab.

   The **Match Models** page appears.
2. Click the name of the match model to which you want to add the Directed AI match rules.
3. On the **Directed AI Match Rules** page, click **Edit**.
4. In the **Rank** column, use the up and down arrows to change the rank of the Directed AI match rules.
5. Click **Save**.

For more information about ranking Directed AI match rules, see “Rank” on page 39.

Edit Directed AI match rules

Before you add Directed AI match rules to a predefined match model, review and edit the rules if required. Also, after you add rules to both predefined models or custom match models created from scratch, you might want to make some changes.

Important: You cannot edit the match strategy and the match criteria for Directed AI match rules that already exist.

1. Open the business entity, and click the **Match** tab.

   The **Match Models** page appears.
2. Click the name of the match model that contains the Directed AI match rules to edit.
3. On the **Directed AI Match Rules** page, click **Edit**, and then click the **Edit** icon beside the rule.

   The **Edit Directed AI Match Rule** dialog box appears.
4. Edit the match fields and properties.

   You can add, edit, and delete match fields.
5. Click **Update**.

   The edited rule appears on the **Directed AI Match Rules** page.
6. Click **Save**.

CHAPTER 7

Configuring Directed AI match rules with relationships

You can configure Directed AI match rules based on relationships and match records. When similar source records are related to another record through a common relationship, the match process considers them as matching records.

For example, consider two users, William Kenneth and William K, who work for Unity Tech. The match process matches the William Kenneth and William K records because they both have the employee relationship with the same organization, Unity Tech.

Consider another scenario where a user, William Kenneth, works for Unity Tech, and another user, Will Kenneth, works for Synergy Solutions. The match process doesn't match the William Kenneth and Will Kenneth records because they aren't related to the same organization through the employee relationship.

Guidelines for configuring Directed AI match rules with relationships

Before you configure Directed AI match rules with relationship match criteria, consider the following guidelines:

- Before you configure relationship matching, ensure that you run a match and merge job on the business entities related with the source records. The match process consolidates duplicate records and makes relationship matching more effective.
- When you configure a Directed AI match rule with relationship-based matching, include additional match fields in the rule to prevent overmatching.

Configure Directed AI match rules with relationship match criteria

You can configure Directed AI match rules to match records based on the relationships that exist between them.

1. On the **Configure Properties and Thresholds** tab, select Relationship as the match criteria.

   The **Relationship** field appears.

2. Click the relationship asset picker.

   The **Select an Asset** dialog box appears.

3. Select a relationship, and click **Select**.
4. Configure the match level, description, match threshold, and click **Next**.

   The **Add Exact Fields** page appears.

5. Add the exact match fields, and click **Next**. To know more about adding an exact match field, see "Step 2. Add required fields" on page 52.

   The **Add Fuzzy Fields** page appears.

   Note: Add at least one exact or fuzzy match field to the Directed AI match rule.

6. Click the **Add Fuzzy Match Field** icon and select a field type.

   A row is added to the **Fuzzy Match Fields** table with the **Comparison Outcome** value set to **Identical**.

7. Select a field name for each field type for matching similar data.

   The field appears in the **Fuzzy Match Fields** table.

8. Click **Next**.

   The **Configure Exact Match Field Properties** page appears.

9. If you added exact match fields, configure the exact match field properties, and click **Add**. To know more about configuring exact match field properties, see "Step 4. Configure field properties" on page 53

   The rule is added to the **Directed AI Match Rules** page.

10. Click **Save**.

CHAPTER 8

Training adaptive AI match models

An Adaptive AI match model is part of the match model. You can train an Adaptive AI match model to predict whether record pairs are a match or not. A trained Adaptive AI match model can analyze data attributes and make match predictions based on its learning.

Before you train an Adaptive AI match model, you need to identify the training data set that is a true representation of your organization's data. The results produced by the Adaptive AI match model are only as good as the training data set the model is trained on.

During the training process, users label record pairs that are presented to them in batches. The labels indicate whether the record pairs are a match or not. The labeled record pairs are the examples that the model learns from to predict whether record pairs are a match.

To train an Adaptive AI match model, use a custom match model that suits your business needs. You create a custom model from a copy of a predefined match model or from scratch.

After you publish a match model, you can define match and merge jobs, and run the jobs. For information about defining and running match and merge jobs, see *Define Jobs*.

Training metrics

When you train an Adaptive AI match model, you need to evaluate the effectiveness of the model. After the model achieves the effectiveness that your business requires, you can complete the training.

To evaluate the effectiveness of the Adaptive AI match model, review the training metrics that appear on the **Adaptive AI Match Model** page.

The following Adaptive AI match model training metrics help answer your questions about model effectiveness:

- "Error matrix" on page 59
- "Accuracy" on page 59
- "Precision" on page 60
- "Recall" on page 60
- "Learning curve" on page 60

Error matrix

An error matrix summarizes the performance of an Adaptive AI match model based on the training data set. Use the matrix to determine how successful the predictions of the model are by understanding the correlation between the actual labels and the predictions of the model.

The following 3 × 3 matrix explains what each cell in an Adaptive AI match model error matrix represents:

-	**Predicted as Match**	**Predicted as Manual Review**	**Predicted as Not a Match**
Labeled as Match	**True positive (TP):** An outcome where the model correctly predicts a record pair as a match.	**False positive (FP):** An outcome where the model incorrectly predicts that a matching record pair requires manual review.	**False negative (FN):** An outcome where the model incorrectly predicts a record pair as not a match.
Labeled as Manual Review	**False positive (FP):** An outcome where the model predicts a record pair that requires manual review as a match.	**True positive (TP):** An outcome where the model correctly predicts that a record pair requires manual review.	**False positive (FP):** An outcome where the model predicts a record pair that requires manual review as not a match.
Labeled as Not a Match	**False positive (FP):** An outcome where the model incorrectly predicts a record pair that's not a match as a match.	**False positive (FP):** An outcome where the model incorrectly predicts a record pair that's not a match requires manual review.	**True negative (TN):** An outcome where the model correctly predicts a record pair as not a match.

The error matrix shows only 30% of the total record pairs that are labeled as a match, manual review, or not a match. For example, if you label 30 record pairs as a match, 20 record pairs as manual review, and 10 record pair as not a match, the error matrix shows the following values:

- 30% of 30 record pairs, which is 9 pairs, predicted as a match.
- 20% of 30 record pairs, which is 6 pairs, predicted as requiring manual review.
- 30% of 10 record pairs, which is 3 pairs, predicted as not a match.

The following image shows an error matrix with record pairs predicted as matches, non-matches, and requiring manual review:

Error Matrix

	Predicted as Match	Predicted as Manual Review	Predicted as Not a Match	Total
Labeled as Match	40% (120)	6.7% (20)	3.3% (10)	50% (150)
Labeled as Manual Review	3.3% (10)	20% (60)	1.7% (5)	25% (75)
Labeled as Not a Match	1.7% (5)	5% (15)	8.3% (25)	25% (75)
Total	45% (135)	31.7% (95)	13.3% (40)	100% (300)

Accuracy

Accuracy measures how well an Adaptive AI match model correctly classifies records as a match, not a match, or requiring manual review.

Accuracy is the percentage of correct predictions out of the total number of predictions.

For example, if the accuracy of an Adaptive AI match model is 95%, it correctly classifies 95% of the record pairs out of the total number of record pairs.

Accuracy = (Total TPs + TN) / (Total TPs + TN + Total FPs + FN) = (Total TPs + TN) / Total number of record pairs

Where TP indicates true positives, TN indicates true negatives, FP indicates false positives, and FN indicates false negatives.

Precision

Precision measures how many of the matches that the Adaptive AI match model predicts are actually correct matches.

Precision is the percentage of matches that the Adaptive AI match model correctly predicts out of the total number of matches it predicts.

For example, if 80% of the record pairs that the Adaptive AI match model identifies as matches are true positives, the precision of the model is 80%.

Precision = Total TPs / (Total TPs + Total FPs) = Total TPs / Total number of predicted matches

Where TP indicates true positives and FP indicates false positives.

Recall

Recall measures how well an Adaptive AI match model can find all the correct matches.

Recall is the percentage of matches that the Adaptive AI match model correctly predicts out of the total number of actual matches.

For example, if the recall of an Adaptive AI match model is 50%, it correctly identifies 50% of record pairs that are a match.

Recall = Total TPs / (Total TPs + FN) = Total TPs / Number of actual matches

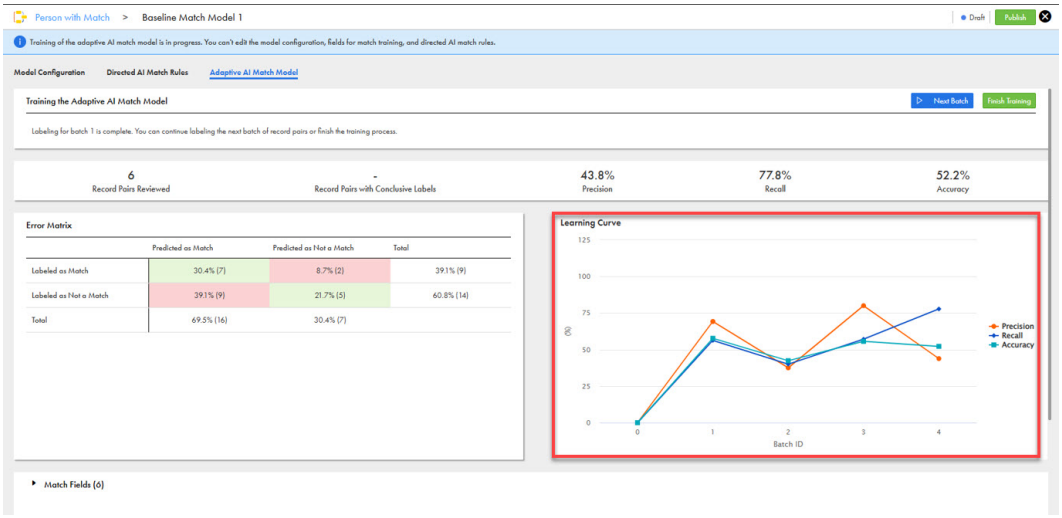
Where TP indicates true positives and FN indicates false negatives.

Learning curve

After you complete the training of record pairs in a batch, you can view a learning curve chart that displays the match training metrics, such as precision, recall, and accuracy.

The learning curve shows the precision, recall, and accuracy for record pairs labeled in each batch individually. You can use these metrics to compare and gain a better understanding of how the model improves as you label record pairs across batches.

The following image shows the training metrics for record pairs labeled across different batches:



Recommendations for selecting training data sets

The accuracy of a match model depends on the quality of the training data set. The training data set provides the examples that the Adaptive AI match model learns from. It is important that the training data set accurately reflects your business scenarios.

To select training data sets for training the Adaptive AI match model, review the following recommendations:

- Start by analyzing the characteristics of your data and the specific match requirements. Understand the data from where you want to select a training data set. Based on the understanding of your data, select a suitable training data set.
- For optimal training results, the data set sample must be similar to the production data. You can manually create a subset of production data by using the field values. For example, use the Country, State, or City code, or a combination of these to create a subset of production data.
- Ensure that the training data set contains the attributes that satisfy your matching requirements. For a relevant data set training, the data set sample must contain attributes same as the production data. For example, if the Phone Number and Email ID attributes are available in the production data, the same attributes must be available in the sample data set that is used for training.
- Ensure that the data sets contain some records that match and some records that do not match so that you have a clear idea of true positives and true negatives when you view the training results.
- For optimal training results, you can avoid duplicating data sets. If a data set has the same name, address, or any other key attributes in more than 25 records, it results in many duplicate data sets. Exclude some of the data to reduce duplicates.
- During the training process, the Adaptive AI match model learns the patterns and characteristics of the data set. Therefore, the training data set must be large enough for the Adaptive AI match model to understand the data patterns for making match predictions. For example, if you have a complex matching scenario, the Adaptive AI match model might need to use a larger number of match fields to effectively make match predictions. The need for a large number of match fields means that the model requires a larger data set.
- A data set must contain a minimum of 10,000 records. The optimal size data set can contain 50,000 to 100,000 records. A data set must contain less than 200,000 records.

Recommendations for training adaptive AI match models

To train an Adaptive AI match model, review the following recommendations:

- Ensure that you are assigned one of the following predefined user roles:
  - Admin
  - Designer
  - MDM Designer
- You can train an Adaptive AI match model with a subset of production data. Consider training a match model in a non-production environment that does not contain the entire production data.
- A match model that you want to use for training must have the same population as the production data.
- During the training process, the training page displays the precision, recall, and accuracy values that indicate the learning of the Adaptive AI match model. Consider reviewing the table in between the batches during the training process.
- When the precision, recall, and accuracy values display a steady increase for an Adaptive AI match model during the training of several batches, it indicates that the Adaptive AI match model is completely trained and you can stop training the Adaptive AI match model.
- For an optimal Adaptive AI match model training, you might want to train the model for at least 20 batches.

Prepare to train an Adaptive AI match model

Before you train an Adaptive AI match model, prepare for the training.

For more information on recommendations for training the Adaptive AI match model, see "Recommendations for training adaptive AI match models" on page 62.

To prepare for training an Adaptive AI match model, perform the following tasks:

"Step 1. Select a training data set" on page 62

"Step 2. Import the training data set" on page 63

"Step 3. Identify a match model" on page 64

"Step 4. Configure match fields" on page 64

Step 1. Select a training data set

Before you train an Adaptive AI match model, select a training data set. Use this data set for training the Adaptive AI match model to predict whether a record pair is a match or not a match. Select a training data set that is comprehensive and accurately reflects your business scenarios.

For more information on recommendations for selecting a training data set, see "Recommendations for selecting training data sets" on page 61.

Note: The Adaptive AI match model training process requires a minimum of 1000 records. The process uses all the data that is available in the Business 360 data store.

Step 2. Import the training data set

You can import a training data set from a CSV file into the Business 360 data store. To import data, you need to use Customer 360. Ensure that your user role has the privileges to import data.

1. Log in to Customer 360.
2. Click **File Import**.

   The **File Import** wizard appears.
3. On the **Upload File** page, drag a CSV source file, or browse and select the source file to upload.
4. Select a source system for the data that you import.
5. In the **File Import Settings** section, verify the system-suggested settings based on your data, and update any setting that is incorrect.

   The following table describes the available import settings:

Import Setting	Description
Delimiter	Character that separates the values in the import file. Select a predefined delimiter or select **Other** to define a custom delimiter.
Text Qualifier	Symbols used in the file to enclose a string.
Encoding Type	Unicode encoding type.
Import Data Starting From Line	Row number in the file from where you want to import data. Use this field to exclude headers. Default is 2.
Contains column headers	Indicates whether the source file includes column headers.
Column Header Row	Header row number in the file.
Regional settings	Indicates whether you want to configure date and time formats.
Date Pattern	Format of the date fields in the file.
Date Time Pattern	Format of the date and time fields in the file. The supported time zone is in UTC (Coordinated Universal Time).
Decimal Separator	Character used as a decimal separator. Default is period (.).
Thousand Separator	Character used as a thousand separator. Default is comma (,).

6. Click **Next**.
7. On the **Map Fields** page, click **Add** to select the target business entity.

   The import process automatically maps the column headers with the fields.
8. To apply a saved mapping, click **Apply a Saved Mapping**, select the mapping, and then click **Apply**.

9. To manually map any field, perform one of the following tasks:
   - Drag a source column to a target field.
   - Select a source column and a corresponding target field, and click **Map Selected**.

   The import process maps the field accordingly.
10. Verify the field mappings and click **Next**.
11. On the **Preview and Import File** page, select one of the following options:

Option	Description
Save as a new mapping	Saves the mapping with the name you specify.
Overwrite an existing mapping	Overwrites a mapping that you select. To apply the changes, click **Apply**.
Don't save	Discards the mapping.

12. Click **Import**.

    The import process starts. You get a notification after the process completes.

Step 3. Identify a match model

Before you train an Adaptive AI match model, review the existing match models. If the existing match model configurations don't meet your business requirements, create a custom model.

The model that you identify might have pre-configured Directed AI match rules and candidate selection criteria. To create a hybrid match model that uses a combination of Directed AI match rules and an Adaptive AI match model, retain the Directed AI match rules. Edit the Directed AI match rules and the candidate selection criteria if required. To create a match model that uses an Adaptive AI match model only, delete any Directed AI match rules that the model contains.

For more information, see Chapter 4, “Configuring match models” on page 29 and Chapter 6, “Configuring Directed AI match rules” on page 50.

Step 4. Configure match fields

After you identify the match model for training, configure the match fields that the Adaptive AI match model requires for the training process.

You can review the match fields before you start training the Adaptive AI match model. The Adaptive AI match model uses the match fields that you configure for the Directed AI match rules, candidate selection criteria, or both. Apart from the existing match fields, you can add additional match fields to train the Adaptive AI match model.

Vew the following details for the match fields that are part of the training:

- Field Name. Name of the match field.

- Field Path. Complete path of the match field. If the field does not have a parent or path, the **Field Path** column does not display any value.
- Internal ID. Unique internal identifier for the match field.
- Used in. Information about where the match field is used. The supported values are Directed AI match rules, candidate selection criteria, and training.
  If you add a match field, the **Used in** column displays **Training** as the default value.

You can delete the match fields that you add. You cannot delete a match field that is part of Directed AI match rules or candidate selection criteria.

Match fields for a pure Adaptive AI match model approach

If you adopt a pure Adaptive AI match model approach, your match model must not contain any Directed AI match rules. The Adaptive AI match model uses the match fields that the candidate section process uses. The match fields that you configure as part of the candidate selection criteria impact the training and determine the accuracy of the Adaptive AI match model.

Match fields for a hybrid Adaptive AI match model approach

If you adopt a hybrid Adaptive AI match model approach based on a set of Directed AI match rules and an Adaptive AI match model, your match model must contain Directed AI match rules and candidate selection criteria. The hybrid model uses the match fields that the Directed AI match rules and candidate section process use. The match fields that you configure impact the training and determine the accuracy of the Adaptive AI match model.

For information about match fields, see "Match fields" on page 41 and "Candidate selection criteria" on page 19.

Training the Adaptive AI match model

Before you can use an Adaptive AI match model for match predictions, you must train and publish the Adaptive AI match model. Before you start training the Adaptive AI match model, you can review and add the match fields that you want to use in the training process. When you start the training process, the training data set is processed to generate record pairs. To train the Adaptive AI match model, label these record pairs that are presented to you in batches.

The training metrics, such as accuracy, precision, and recall, are calculated after a batch of record pairs is labeled. When you reach the desired level of accuracy, precision, and recall, complete the training process.

Note: The training process uses all the data that is available in the MDM SaaS data store.

To train an Adaptive AI match model, you must label the record pairs from the training data set. You can label the record pairs in multiple batches, up to a maximum of 30 batches. Each batch might contain record pairs in multiple sections that are grouped based on exact match fields. For example, you might find a section for record pairs that match exactly based on the Address field and another section for record pairs that match exactly based on the Country field.

Record pairs without exact matches are also grouped in a separate section. You must label 30 record pairs with conclusive labels, Match or Not a Match, within each batch. You can complete labeling a batch with 30 conclusive pairs or continue labeling up to 90 pairs to improve match accuracy.

When you create a match model, the match process recommends match fields from the candidate selection criteria and Directed AI match rules for inclusion in the Adaptive AI match model training. You can also modify the configuration of the match model even after you start training its Adaptive AI match model. You can add or remove match fields from existing Directed AI match rules or add new Directed AI match rules to

the match model. The match process then displays the newly added match fields as recommendations for inclusion in the Adaptive AI match model training.

Additionally, if you modify the following settings of a match model during or after training the Adaptive AI match model, the match process discards the training and requires you to retrain the model from the beginning:

- Default population
- Candidate selection criteria
- Match fields in the Adaptive AI match model

Note: If you upgrade from a release prior to April 2026, match fields previously added to your Adaptive AI match models are excluded from the match process after the upgrade. To restore the match fields, create a new version of the match model and add the fields to it.

Label record pairs for training an Adaptive AI match model

Label record pairs in bulk within a section or individually after assessing them. After labeling the required number of record pairs in all batches, you can complete the training.

1. Open a business entity, and click the **Match** tab.

   The **Match Models** page appears.
2. Click the match model that was identified for training the Adaptive AI match model, and click the **Adaptive AI Match Model** tab.

   The **Adaptive AI Match Model** page appears.
3. In the **Match Fields** section, review the match fields.

   The fields that the Directed AI match rules and candidate selection criteria use appear as recommended match fields.
4. To add the recommended match fields, perform the following steps:
   a. Click the recommended match fields that appear as hyperlinks.
   b. From the **Add Recommended Match Fields** dialog box, select the match fields to add, and click **Add**.
5. To remove any match field, Select the field that you want to remove, and click **Actions** > **Remove**.
6. To add additional match fields, click **Manage Fields**.
7. In the **Manage Fields** dialog box, select the fields that you want to add, and click **OK**.

   The selected fields are added for training the model.
8. After you review and add the match fields, click **Start Training**.

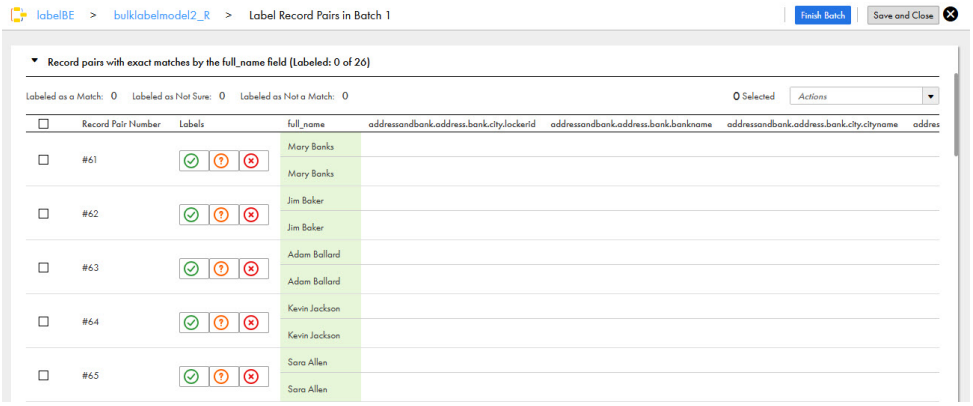
   The training process starts preparing the data for training.
9. After the training data preparation completes, click **Start Labeling**.

   The **Label Record Pairs in Batch** page appears. The record pairs are grouped into sections based on exact match fields that matched them.

10. To label record pairs, perform the following steps:

    a. Expand a section that contains the record pairs to label.

    Each section contains up to 90 record pairs.
    The following image shows a section that lists record pairs matched exactly by the Full Name field:



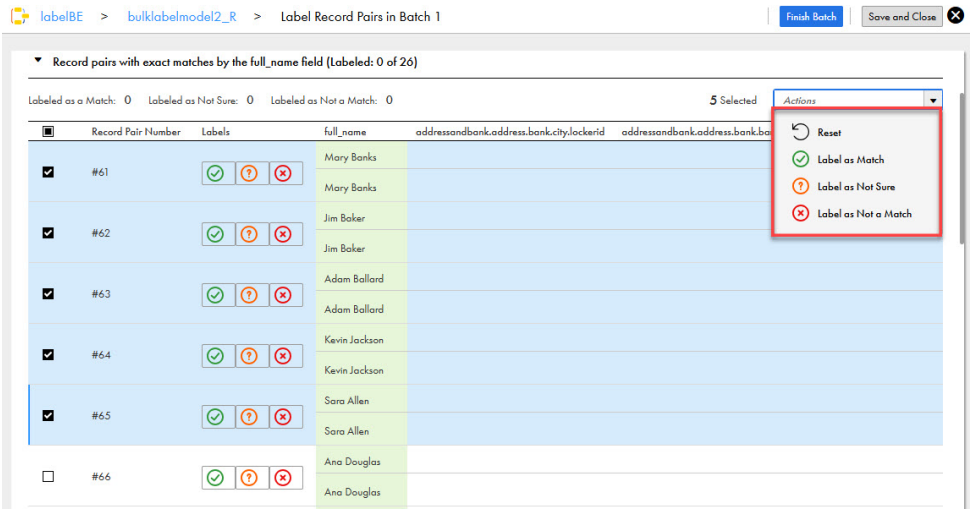
    b. Select a record pair and label it with one of the following labels:

Label	Description
Match	A conclusive label. Indicates that the records in the pair are the same.
Not a Match	A conclusive label. Indicates that the records in the pair are different.
Requires Manual Review	A conclusive label. Indicates that the record pairs require manual review by a data steward.
Skip	An inconclusive label. Indicates that you want to exclude record pairs from labeling. The record pairs with the Skip label don't appear for labeling again within the same batch during the training process.

    c. Optionally, if you are confident about the record pairs, you can select multiple record pairs within a section and label them in bulk.

    d. If you selected multiple record pairs, click **Actions**.

    The following image shows multiple record pairs selected within a section for bulk labeling:

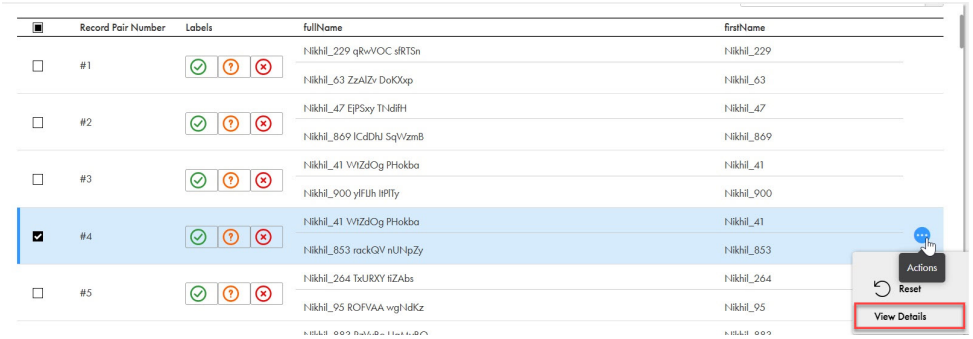


e. Select a label.

f. To reset the label assigned to a record pair, select the record pair, and click **Actions** > **Reset**.

The label assigned to the selected record pair is reset and the record pair is set to participate in the training process again.

g. To view the details of a record pair, select the record pair and click **Actions** > **View Details**.



h. Optionally, you can continue labeling the record pair without navigating to the **Label Record Pairs in Batch** page. To label the record pair in the **Details of Record Pair** dialog box, select a label.

i. To label the next record pair, click **Next**.

j. Select another section and continue labeling.

After you select conclusive labels for 30 record pairs, the labeling for the batch is complete. However, you can label up to 90 record pairs to improve the match accuracy.

k. To review the labeled record pairs, click **Previous** and **Next**.

11. To complete labeling the batch, click **Finish Batch**.

The accuracy of the Adaptive AI match model is calculated.

12. To label the next batch of record pairs, click **Next Batch**, and label record pairs by repeating step 10.

13. To pause labeling, on the **Label Record Pairs in Batch** page, click **Save and Close**.

14. To resume labeling, on the **Match Training** page, click **Continue Labeling**, and label record pairs by repeating step 10.

15. After you complete labeling the required number of batches and reach an acceptable accuracy, precision, and recall, click **Finish Training**.

    The training process completes and saves the Adaptive AI match model as part of the match model.

Reset an Adaptive AI match model training

You can reset the training of an Adaptive AI match model during or after training if the training fails to produce the desired results.

When you reset the training of an Adaptive AI match model, the match process discards the labeled record pairs and training metrics of the current version. You can then retrain the model. You can reset the training when the match model is in the draft state.

1. Open the draft match model that contains the Adaptive AI match model training you want to reset.
2. Open the **Adaptive AI Match Model** page.
3. To reset the training of the Adaptive AI match model, click **Reset Training**.

   A **Warning** dialog box appears.
4. Click **Reset**.
5. In the **Match Fields** section, review the match fields.

   The fields that the Directed AI match rules and candidate selection criteria use appear as recommended match fields.
6. To add the recommended match fields, perform the following steps:
   a. Click the recommended match fields that appear as hyperlinks.
   b. From the **Add Recommended Match Fields** dialog box, select the match fields to add, and click **Add**.
7. To remove any match field, Select the field that you want to remove, and click **Actions** > **Remove**.
8. To add additional match fields, click **Manage Fields**.
9. In the **Manage Fields** dialog box, select the fields that you want to add, and click **OK**.

   The selected fields are added for training the model.
10. After you review and add the match fields, click **Start Training**.

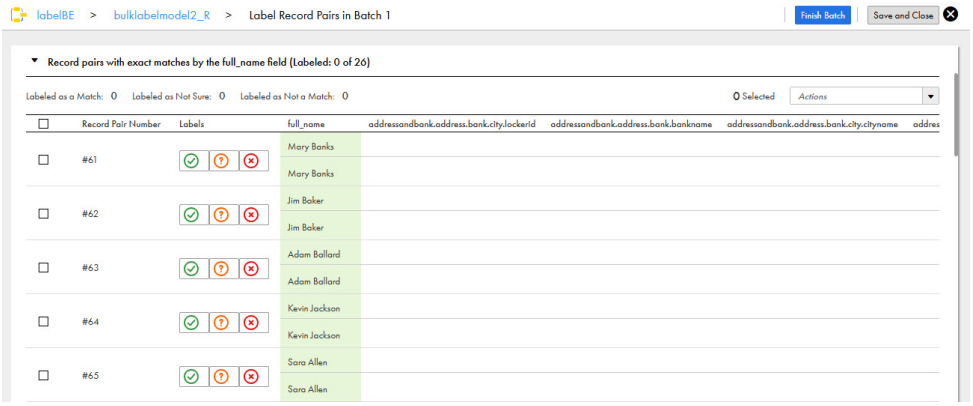
    The training process starts preparing the data for training.
11. After the training data preparation completes, click **Start Labeling**.

    The **Label Record Pairs in Batch** page appears. The record pairs are grouped into sections based on exact match fields that matched them.

12. To label record pairs, perform the following steps:

    a. Expand a section that contains the record pairs to label.

    Each section contains up to 90 record pairs.
    The following image shows a section that lists record pairs matched exactly by the Full Name field:



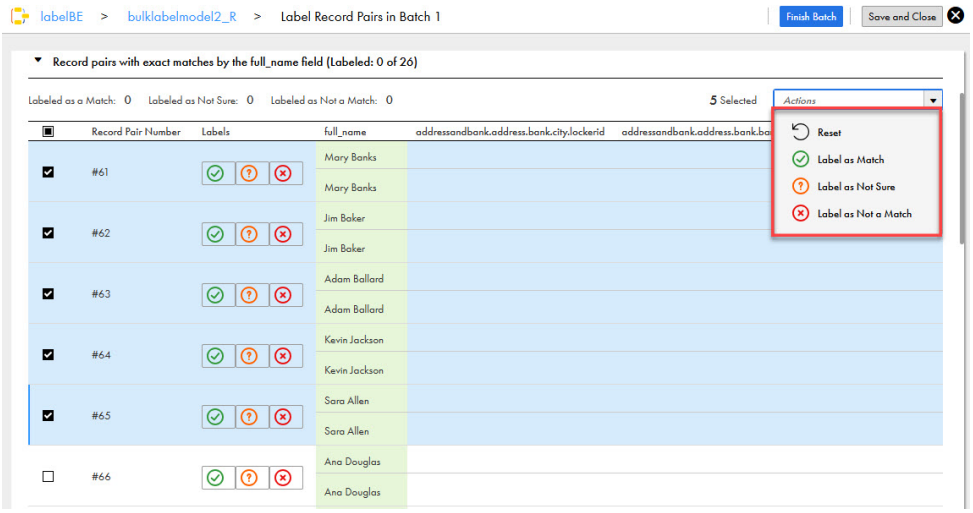
    b. Select a record pair and label it with one of the following labels:

Label	Description
Match	A conclusive label. Indicates that the records in the pair are the same.
Not a Match	A conclusive label. Indicates that the records in the pair are different.
Requires Manual Review	A conclusive label. Indicates that the record pairs require manual review by a data steward.
Skip	An inconclusive label. Indicates that you want to exclude record pairs from labeling. The record pairs with the Skip label don't appear for labeling again within the same batch during the training process.

    c. Optionally, if you are confident about the record pairs, you can select multiple record pairs within a section and label them in bulk.

    d. If you selected multiple record pairs, click **Actions**.

    The following image shows multiple record pairs selected within a section for bulk labeling:

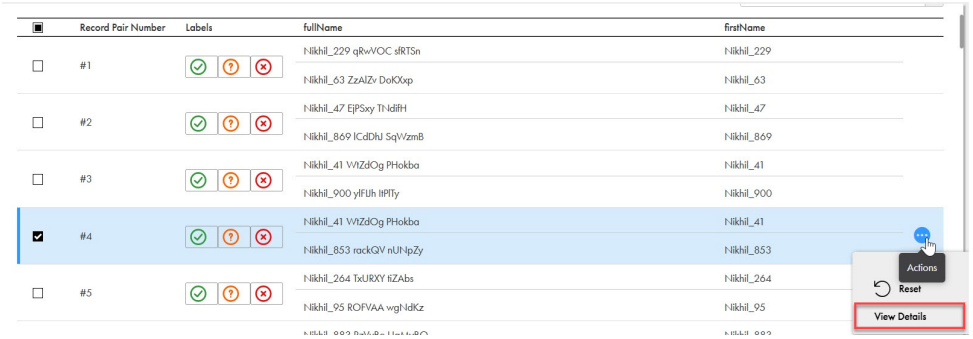


e. Select a label.

f. To reset the label assigned to a record pair, select the record pair, and click **Actions** > **Reset**.

The label assigned to the selected record pair is reset and the record pair is set to participate in the training process again.

g. To view the details of a record pair, select the record pair and click **Actions** > **View Details**.



h. Optionally, you can continue labeling the record pair without navigating to the **Label Record Pairs in Batch** page. To label the record pair in the **Details of Record Pair** dialog box, select a label.

i. To label the next record pair, click **Next**.

j. Select another section and continue labeling.

After you select conclusive labels for 30 record pairs, the labeling for the batch is complete. However, you can label up to 90 record pairs to improve the match accuracy.

k. To review the labeled record pairs, click **Previous** and **Next**.

13. To complete labeling the batch, click **Finish Batch**.

The accuracy of the Adaptive AI match model is calculated.

14. To label the next batch of record pairs, click **Next Batch**, and label record pairs by repeating step 12.

15. To pause labeling, on the **Label Record Pairs in Batch** page, click **Save and Close**.

16. To resume labeling, on the **Match Training** page, click **Continue Labeling**, and then label record pairs by repeating step 12.

17. After you complete labeling the required number of batches and reach an acceptable accuracy, precision, and recall, click **Finish Training**.

    The training process completes and saves the Adaptive AI match model as part of the match model.

Keyboard shortcuts

When you view the details of a record pair, you can use keyboard shortcuts to label it and also navigate to other pairs in a batch. You can use the keyboard shortcuts in the **Details of Record Pair** dialog box.

The following table lists the available keyboard shortcuts:

Action	Shortcut Key
Label a record pair as match.	M
Label a record pair as not sure.	U
Label a record pair as not a match.	N
Reset a labeled record pair.	R
Navigate to previous pair.	Left arrow
Navigate to next pair.	Right arrow

CHAPTER 9

Publishing match models

To use a match model for the match and merge process, publish the model after you complete configuring it. After you publish a model, you can define match and merge jobs that use the model. You cannot delete or change a published model back to the draft state.

Important: You must configure and publish the match model after you configure the data model, but before the ingress job is run for the initial data load. If the match model configuration is published after the initial data load, the data that is already in the Business 360 data store is not considered by the match model and does not undergo matching.

If you publish a match model with changes to the candidate selection criteria, the model is applied to the data that loads after you publish the model.

Publish a match model

To use a match model for the match and merge process, you must publish the model. You can publish the model after you complete configuring the match model properties and the Directed AI match rules.

1. Open the business entity, and click the **Match** tab.

   The **Match Models** page appears.
2. Click the name of the match model that you want to publish.
3. On the page that opens, click **Publish**.

   A **Warning** dialog box appears.
4. Based on the match model configurations, review the warning, and click one of the following actions:
   - **Publish**. Publishes the match model configuration. The match model is available for the generate search indexes or match keys job.
   - **Publish and Regenerate Match Keys**. Publishes the match configuration and then generates match keys for the existing and new records.
     When you click **Publish and Regenerate Match Keys**, a generate search indexes or match keys job is created. You can view the status of the job on the **My Jobs** page.

   On the **Match Models** page, the status of the match model is changed from draft to published.
5. **Next step:** Define match and merge jobs, and run the jobs.

   For information about defining and running match and merge jobs, see *Define Jobs*.

Additional training for Adaptive AI match models

After you publish a match model, you can continue training its Adaptive AI match model to enhance its accuracy and precision of the Adaptive AI match model.

When you continue training an Adaptive AI match model, the match process creates a new version of the match model and updates the Adaptive AI match model based on the outcome of record pairs that you label. After you complete the training, the Adaptive AI match model updates its rules and recalculates the training metrics based on the updated rules.

Continue training an Adaptive AI match model

After you publish a match model, you can continue training its Adaptive AI match model.

1. Open the published match model that contains the trained Adaptive AI match model.
2. Open the **Adaptive AI Match Model** page.
3. To continue training the Adaptive AI match model, click **Additional Training**.

   Note: When you import assets from existing organizations into new organizations, you can't view the **Additional Training** button on the Adaptive AI match model page for existing match models in the imported organizations.
4. In the **Confirm Match Training** dialog box, click **Start**.
5. In the **Match Fields** section, review the match fields.

   The fields that the Directed AI match rules and candidate selection criteria use appear as recommended match fields.
6. To add the recommended match fields, perform the following steps:
   a. Click the recommended match fields that appear as hyperlinks.
   b. From the **Add Recommended Match Fields** dialog box, select the match fields to add, and click **Add**.
7. To remove any match field, Select the field that you want to remove, and click **Actions** > **Remove**.
8. To add additional match fields, click **Manage Fields**.
9. In the **Manage Fields** dialog box, select the fields that you want to add, and click **OK**.

   The selected fields are added for training the model.
10. After you review and add the match fields, click **Start Training**.

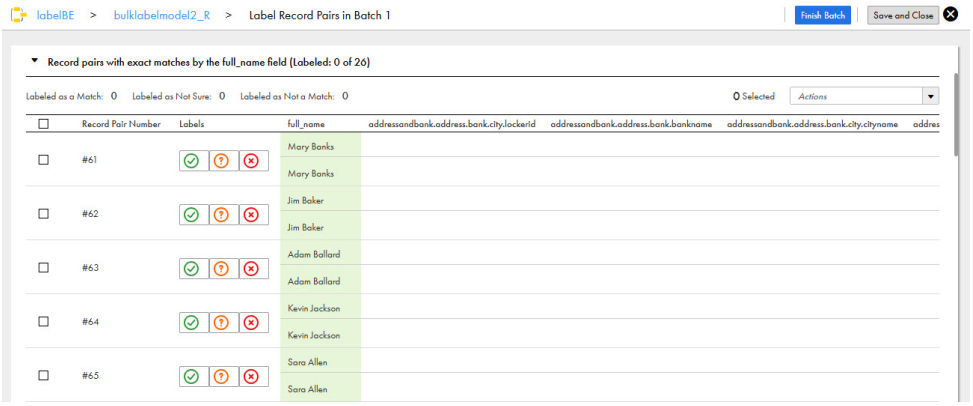
    The training process starts preparing the data for training.
11. After the training data preparation completes, click **Start Labeling**.

    The **Label Record Pairs in Batch** page appears. The record pairs are grouped into sections based on exact match fields that matched them.

12. To label record pairs, perform the following steps:

    a. Expand a section that contains the record pairs to label.

    Each section contains up to 90 record pairs.
    The following image shows a section that lists record pairs matched exactly by the Full Name field:



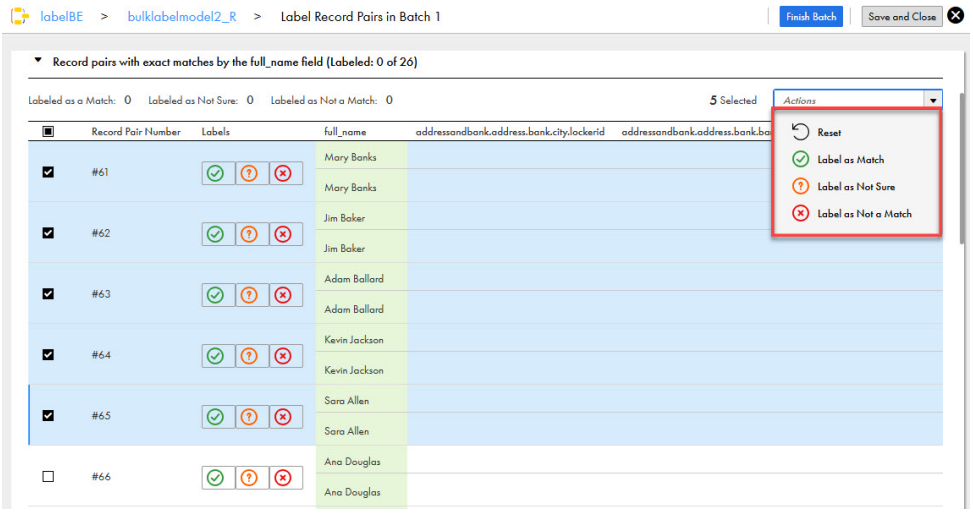
    b. Select a record pair and label it with one of the following labels:

Label	Description
Match	A conclusive label. Indicates that the records in the pair are the same.
Not a Match	A conclusive label. Indicates that the records in the pair are different.
Requires Manual Review	A conclusive label. Indicates that the record pairs require manual review by a data steward.
Skip	An inconclusive label. Indicates that you want to exclude record pairs from labeling. The record pairs with the Skip label don't appear for labeling again within the same batch during the training process.

    c. Optionally, if you are confident about the record pairs, you can select multiple record pairs within a section and label them in bulk.

    d. If you selected multiple record pairs, click **Actions**.

    The following image shows multiple record pairs selected within a section for bulk labeling:

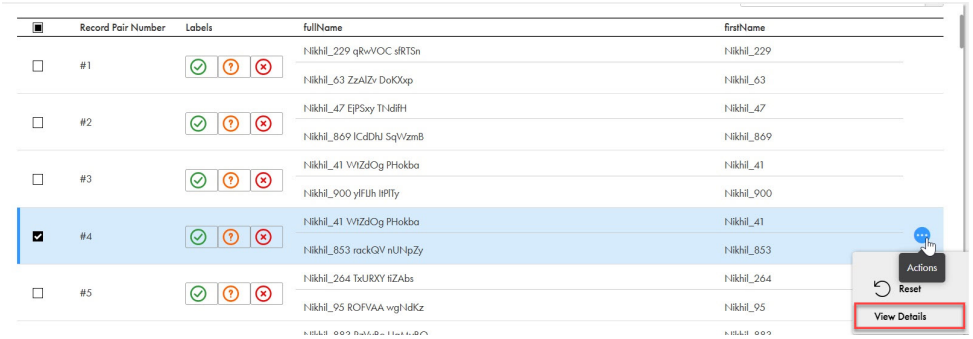


e. Select a label.

f. To reset the label assigned to a record pair, select the record pair, and click **Actions** > **Reset**.

The label assigned to the selected record pair is reset and the record pair is set to participate in the training process again.

g. To view the details of a record pair, select the record pair and click **Actions** > **View Details**.



h. Optionally, you can continue labeling the record pair without navigating to the **Label Record Pairs in Batch** page. To label the record pair in the **Details of Record Pair** dialog box, select a label.

i. To label the next record pair, click **Next**.

j. Select another section and continue labeling.

After you select conclusive labels for 30 record pairs, the labeling for the batch is complete. However, you can label up to 90 record pairs to improve the match accuracy.

k. To review the labeled record pairs, click **Previous** and **Next**.

13. To complete labeling the batch, click **Finish Batch**.

The accuracy of the Adaptive AI match model is calculated.

14. To label the next batch of record pairs, click **Next Batch**, and label record pairs by repeating step 12.

15. To pause and resume labeling later on the Match Training page click **Continue Labeling**, and then label record pairs by repeating step 12.

16. After you complete labeling the required number of batches and reach an acceptable accuracy, precision, and recall, click **Finish Training**.

The training process completes and saves the Adaptive AI match model as part of the match model.

Retrain an Adaptive AI match model

You can retrain an Adaptive AI match model as a new version if the training fails to produce the desired results after you publish the match model.

When you retrain an Adaptive AI match model, the match process creates a draft version of the match model without including the Adaptive AI match model. You can then use the draft version of the match model and start training the Adaptive AI match model from the beginning.

Additionally, the match process discards the labeled record pairs and training metrics from all previous versions of the published match model. For example, if you retrain version 2 of an Adaptive AI match model as a new version, the match process creates a draft version called version 3 and discards training metrics and labeled record pairs from versions 1 and 2.

Retrain as a new version

After you publish a match model, you can retrain its Adaptive AI match model as a new version.

1. Open the published match model that contains the trained Adaptive AI match model to retrain.
2. Open the **Adaptive AI Match Model** page.
3. To retrain the Adaptive AI match model as a new version, click **Train as a New Version**.

   Note: When you import assets from existing organizations into new organizations, you can't view the **Train as a New Version** button on the Adaptive AI match model page for existing match models in the imported organizations.

   A **Warning** dialog box appears.
4. Click **Create**.

   The **Adaptive AI Match Model** page appears.
   The match process creates a draft version of the match model without the Adaptive AI match model. Additionally, the match process discards labeled record pairs and training metrics from all previous versions of the published match model.
5. In the **Match Fields** section, review the match fields.

   The fields that the Directed AI match rules and candidate selection criteria use appear as recommended match fields.
6. To add the recommended match fields, perform the following steps:
   a. Click the recommended match fields that appear as hyperlinks.
   b. From the **Add Recommended Match Fields** dialog box, select the match fields to add, and click **Add**.
7. To remove any match field, Select the field that you want to remove, and click **Actions** > **Remove**.
8. To add additional match fields, click **Manage Fields**.
9. In the **Manage Fields** dialog box, select the fields that you want to add, and click **OK**.

   The selected fields are added for training the model.
10. After you review and add the match fields, click **Start Training**.

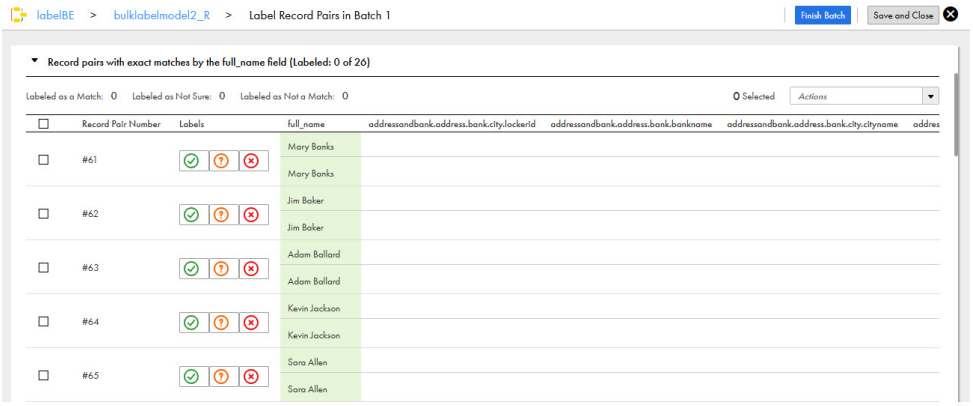
    The training process starts preparing the data for training.
11. After the training data preparation completes, click **Start Labeling**.

    The **Label Record Pairs in Batch** page appears. The record pairs are grouped into sections based on exact match fields that matched them.

12. To label record pairs, perform the following steps:

    a. Expand a section that contains the record pairs to label.

    Each section contains up to 90 record pairs.
    The following image shows a section that lists record pairs matched exactly by the Full Name field:



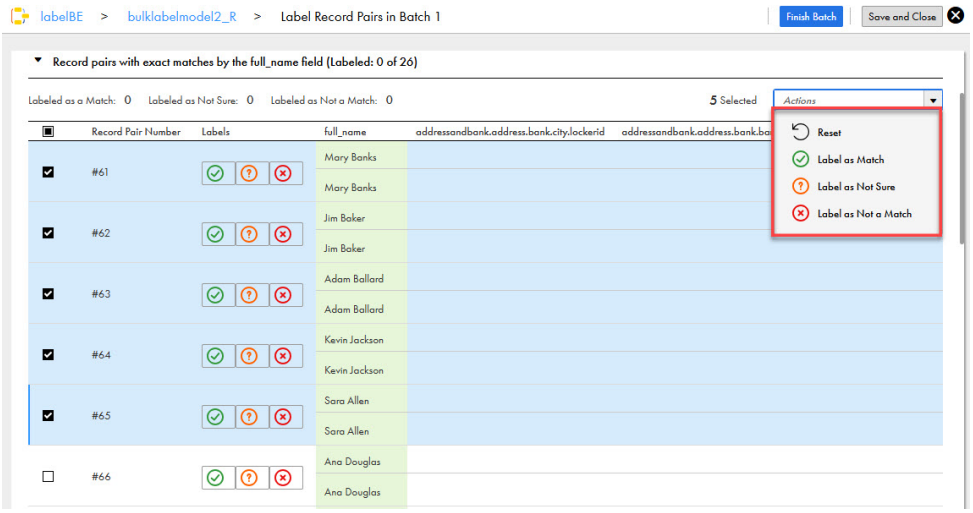
    b. Select a record pair and label it with one of the following labels:

Label	Description
Match	A conclusive label. Indicates that the records in the pair are the same.
Not a Match	A conclusive label. Indicates that the records in the pair are different.
Requires Manual Review	A conclusive label. Indicates that the record pairs require manual review by a data steward.
Skip	An inconclusive label. Indicates that you want to exclude record pairs from labeling. The record pairs with the Skip label don't appear for labeling again within the same batch during the training process.

    c. Optionally, if you are confident about the record pairs, you can select multiple record pairs within a section and label them in bulk.

    d. If you selected multiple record pairs, click **Actions**.

    The following image shows multiple record pairs selected within a section for bulk labeling:

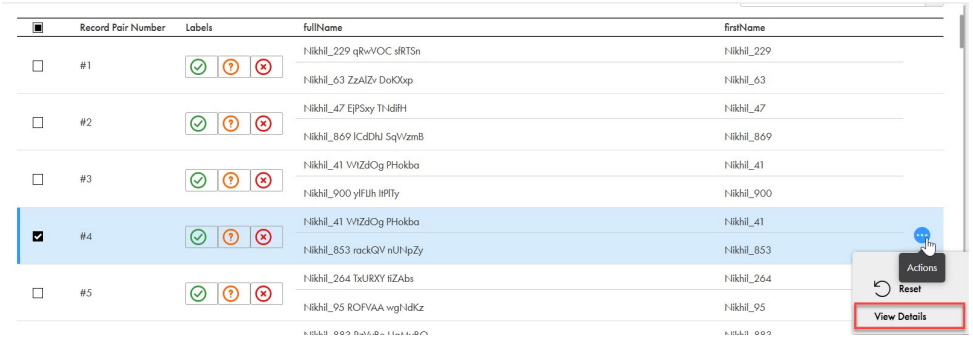


e. Select a label.

f. To reset the label assigned to a record pair, select the record pair, and click **Actions** > **Reset**.

The label assigned to the selected record pair is reset and the record pair is set to participate in the training process again.

g. To view the details of a record pair, select the record pair and click **Actions** > **View Details**.



h. Optionally, you can continue labeling the record pair without navigating to the **Label Record Pairs in Batch** page. To label the record pair in the **Details of Record Pair** dialog box, select a label.

i. To label the next record pair, click **Next**.

j. Select another section and continue labeling.

After you select conclusive labels for 30 record pairs, the labeling for the batch is complete. However, you can label up to 90 record pairs to improve the match accuracy.

k. To review the labeled record pairs, click **Previous** and **Next**.

13. To complete labeling the batch, click **Finish Batch**.

The accuracy of the Adaptive AI match model is calculated.

14. To label the next batch of record pairs, click **Next Batch**, and label record pairs by repeating step 12.

15. To pause and resume labeling later on the Match Training page click **Continue Labeling**, and then label record pairs by repeating step 12.

16. After you complete labeling the required number of batches and reach an acceptable accuracy, precision, and recall, click **Finish Training**.

The training process completes and saves the Adaptive AI match model as part of the match model.

CHAPTER 10

Setting the default match model for the search match API

The search match API searches for matching records based on a match model. You can set any published match model as a default match model for the search match API. After you set the default match model, the search match API uses it for matching records.

1. On the **Match Models** page, click the **Actions** icon beside the name of the match model that you want to set as the default model for the search match API.
2. Select **Set as Default for Search Match API**.

   The match model is set as the default match model for the search match API.

INDEX

A

adaptive AI match model
  about 58
  accuracy 59
  build 58
  configure 58
  precision 60
  recall 60
  set up 58
  training 58
  training metrics 59, 60

B

business entity field
  field type 20

C

candidate selection
  field name 20

D

declarative match rule
  configuring 50, 54
  editing 54
  execution order 54
  ranking 54
  reviewing data 50
declarative rule
  configuring 50, 54
  description 35
  editing 54
  execution order 54
  field type 42
  match criteria 38
  match level 39
  match strategies 36
  merge strategies 36
  properties 35
  ranking 39, 54
  reviewing data 50
directed AI match rule
  adding 51
  configuring 51

I

Informatica Global Customer Support
  contact information 7
Informatica Intelligent Data Management Cloud
  web site 6

M

machine learning model
  before you train a model 62
  error matrix 59
  prepare for training 62
  training metrics 58, 59
maintenance outages 7
match
  configure 8
  configuring 29
  criteria 38
  field type 20, 42
  model name 12
  model objective 12
  population 13
  strategies 36
match candidate
  search level 23
  selection criteria 19
match field
  about 41
  field type 42
match key
  key generation level 22
match level 39
match model
  about 29
  candidate search level 23
  configuring 29, 30
  creating 29, 31
  creating a copy 30
  creating from a copy 29
  creating from scratch 31
  customizing 30
  field type 20
  for predefined and custom business entities 31
  for predefined business entities 29
  key generation level 22
  name 12
  objective 12
  population 13
  properties 12
  publishing 73
  reviewing 29
match rule
  adding 51
  configuring 50, 51, 54
  description 35
  editing 54
  execution order 54
  properties 35

match rule *(continued)*
ranking 39, 54
reviewing data 50
match training
metrics 58
matching records
prerequisites 8
merge
configure 8
configuring 29
strategies 36
thresholds 36
merging records
prerequisites 8
model name
match 12
model objective
match 12

P

population
match 13

S

status
IDMC 7
system status 7

T

training data set
importing 63
selecting 62
trust site
description 7

U

upgrade notifications 7

W

web site 6